

063-0605-00L : Computational Structural Design 1

# Computational Graphic Statics

Dr. Juney Lee & Dr. Lluís Enrique

Autumn Semester 2021

**ETH** zürich

**DARCH**

41

**BRG**

## Week 9

Friday, December 3rd

Hour 1

09:45 – 09:50

- Recap of Exercise 5

09:50 – 10:30

- **Lecture 6 : 3D graphic statics**

10:30 – 10:45

- Break

Hour 2

10:45 – 11:35

- **Tutorial 6 : Polyhedral 3D Graphic Statics (3GS)**

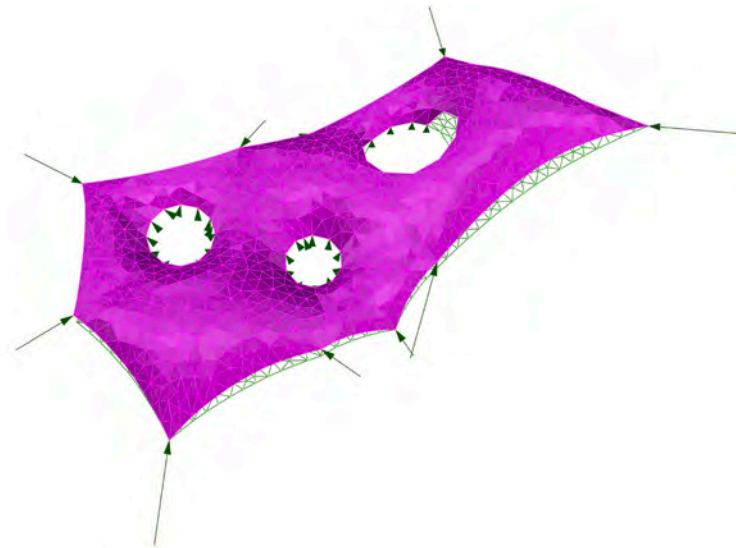
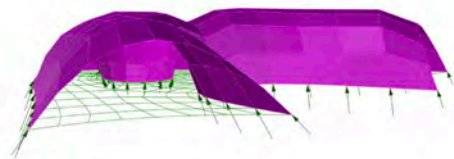
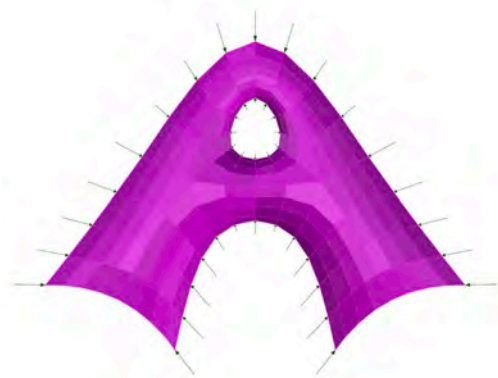
11:30 – 11:45

- Break

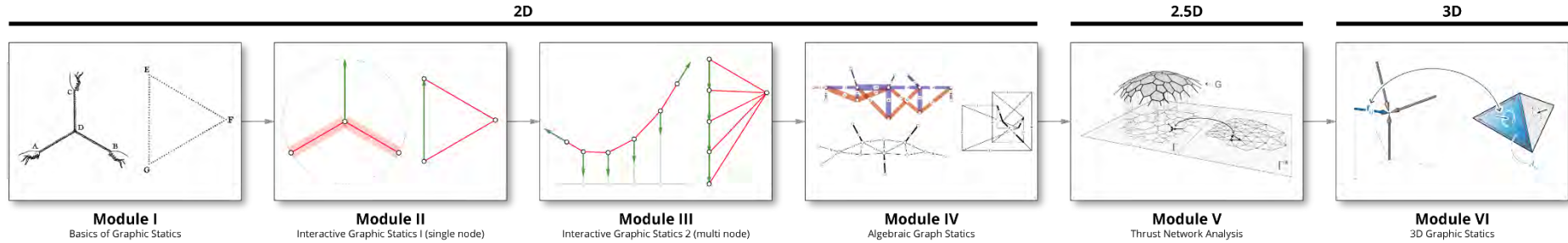
Hour 3

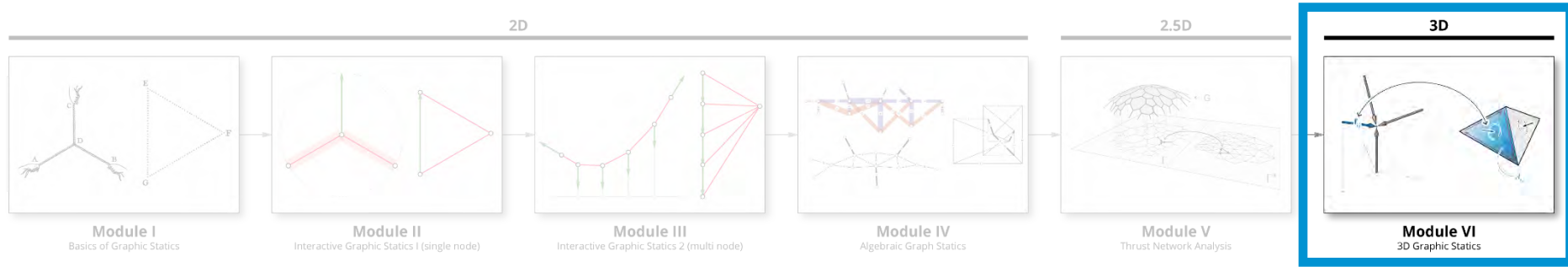
11:45 – 12:30

- **Exercise 6 : Polyhedral 3D Graphic Statics (3GS)**

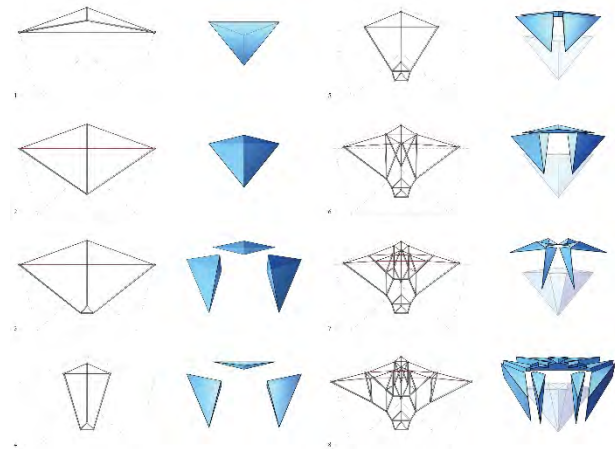
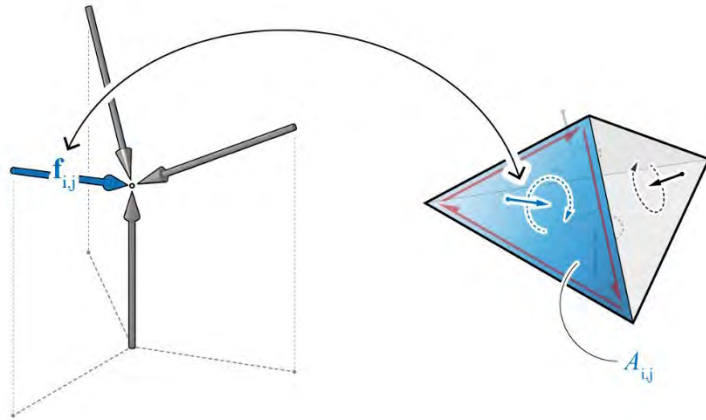


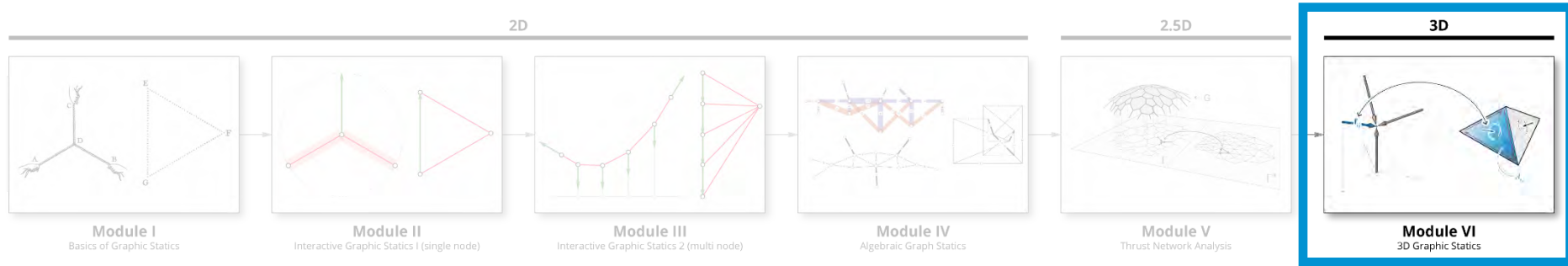
| Structure Type(s)    |                    |     | Topic        | Week            | Day                                                                                           | Topic                                                      | Instructor(s)                   |
|----------------------|--------------------|-----|--------------|-----------------|-----------------------------------------------------------------------------------------------|------------------------------------------------------------|---------------------------------|
| 1. planar structures | cables & arches    | I   | Introduction | Week 1 (9/24)   | 1                                                                                             | Lecture 1 Introduction to graphic statics                  | Dr. Juney Lee                   |
|                      |                    |     |              | 2               | Tutorial 1 Basics of graphic statics on "blackboard" Introduction to Rhino + Grasshopper (GH) | Dr. Lluís Enríque<br>Lotte Aldinger                        |                                 |
|                      |                    |     |              | 3               | Work session 1 Exercise 1: Reciprocal diagrams in Rhino + GH basics                           | Instructors                                                |                                 |
|                      | 2D GS (procedural) | II  |              | Week 2 (10/1)   | 1                                                                                             | Lecture 2 Quick recap of Exercise 1                        | Lotte Aldinger                  |
|                      |                    |     |              | 2               | Tutorial 2 Algorithmic design & thinking                                                      | Dr. Juney Lee                                              |                                 |
|                      |                    |     |              | 3               | Tutorial 2 Single node bridge in GH (form to force)                                           | Lotte Aldinger                                             |                                 |
|                      |                    |     |              | Week 3 (10/8)   | 1                                                                                             | Work session 2 Exercise 2: Single node bridge with GH      | Lotte Aldinger + Instructors    |
|                      |                    |     |              | 2               |                                                                                               |                                                            |                                 |
|                      |                    |     |              | 3               |                                                                                               |                                                            |                                 |
|                      |                    | III |              | Week 4 (10/15)  | 1                                                                                             | Lecture 3 Quick recap of Exercise 2                        | Lotte Aldinger                  |
|                      |                    |     |              | 2               | Tutorial 3 Computational graphic statics                                                      | Dr. Juney Lee                                              |                                 |
|                      |                    |     |              | 3               | Tutorial 3 Multi-node bridge in Grasshopper                                                   | Dr. Lluís Enríque                                          |                                 |
|                      |                    |     |              | Week 5 (10/22)  | 1                                                                                             | Work session 3 Exercise 3: Multi-node bridge with GH       | Dr. Lluís Enríque + Instructors |
|                      |                    |     |              | 2               |                                                                                               |                                                            |                                 |
|                      |                    |     |              | 3               |                                                                                               |                                                            |                                 |
| Week 6 seminar week  |                    |     |              |                 |                                                                                               |                                                            |                                 |
| trusses              | 2D GS (AGS)        | IV  |              | Week 7 (11/5)   | 1                                                                                             | Lecture 4 Quick recap of Exercise 3                        | Dr. Lluís Enríque               |
|                      |                    |     |              | 2               | Tutorial 4 Algebraic graph statics (AGS)                                                      | Dr. Juney Lee                                              |                                 |
|                      |                    |     |              | 3               | Tutorial 4 compas_agr + IGS (Interactive Graphic Statics) plugin                              | Ricardo Avelino                                            |                                 |
|                      |                    |     |              | Week 8 (11/12)  | 1                                                                                             | Work session 4 Exercise 4: Truss problems with IGS         | Ricardo Avelino + Instructors   |
|                      |                    |     |              | Week 9 (11/19)  | 1                                                                                             | Lecture 5 Quick recap of Exercise 4                        | Ricardo Avelino                 |
|                      |                    |     |              | 2               | Tutorial 5 Thrust Network Analysis (TNA)                                                      | Dr. Juney Lee                                              |                                 |
|                      |                    |     |              | 3               | Tutorial 5 compas_tna + RV2 (RhinoVAULT 2) plugin                                             | Dr. Juney Lee                                              |                                 |
|                      |                    |     |              | Week 10 (11/26) | 1                                                                                             | Work session 5 Exercise 5: Shell design exercises with RV2 | Dr. Juney Lee + Instructors     |
|                      |                    |     |              | Week 11 (12/3)  | 1                                                                                             | Lecture 6 Quick recap of Exercise 5                        | Dr. Juney Lee                   |
|                      |                    |     |              | 2               | Tutorial 6 3D graphic statics (3GS)                                                           | Dr. Juney Lee                                              |                                 |
|                      |                    |     |              | 3               | Tutorial 6 compas_3gs + 3GS (3D graphic statics) plugin                                       | Dr. Juney Lee                                              |                                 |
|                      |                    |     |              | Week 12 (12/10) | 1                                                                                             | Lecture 7 Pt I. Outlook on computational graphic statics   | Dr. Juney Lee                   |
|                      |                    |     |              | 2               | Q/A                                                                                           | Discussions, feedback, evaluations, etc.                   |                                 |





## Module VI · 3D graphic statics





## Module VI · 3D graphic statics



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Computational Graphic Statics

# Lecture 6

## 3D Graphic Statics

Friday, December 3rd, 2021

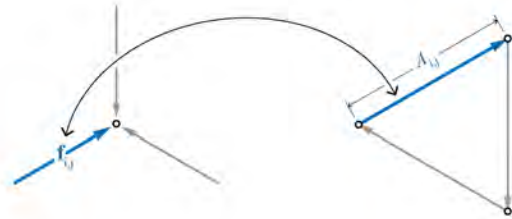
Dr. Juney Lee

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DESIGN RESEARCH CENTER

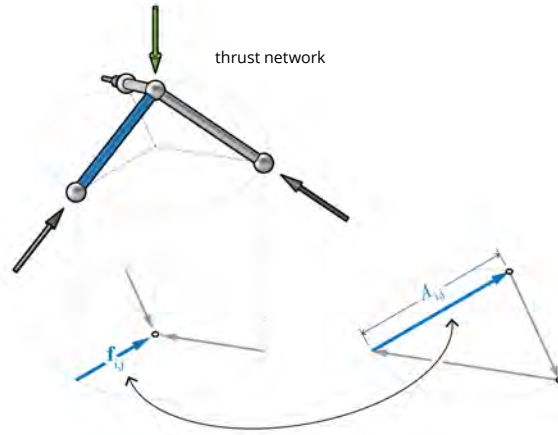
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form diagram

force diagram

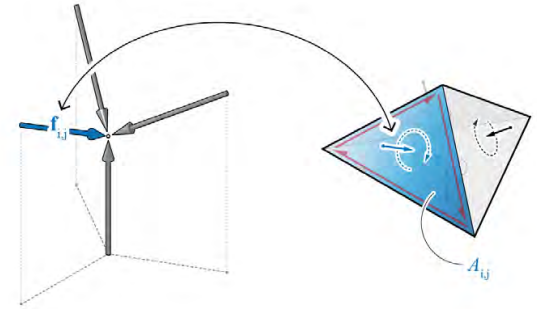
2 D



form diagram

force diagram

2.5 D



form diagram

force diagram

3 D

[ 92 ]

**XVII. Principle of the Equilibrium of Polyhedral Frames.**  
*By W. J. MACQUORN RANKINE, C.E., LL.D., F.R.S.S.L.&E.\**

**T**HE following theorem is the extension to polyhedral frames of a principle which is proved for polygonal frames in 'A Manual of Applied Mechanics,' art. 150.

**THEOREM.**—If planes diverging from a point or line be drawn normal to the lines of resistance of the bars of a polyhedral frame, then the faces of a polyhedron whose edges lie in those diverging planes (in such a manner that those faces, together with the diverging planes which contain their edges, form a set of contiguous diverging pyramids or wedges) will represent, and be normal to, a system of forces which, being applied to the summits of the polyhedral frame, will balance each other—each such force being applied to the summit of meeting of the bars whose lines of resistance are normal to the set of diverging planes that enclose that face of the polyhedron of forces which represents and is normal to the force in question. Also, the areas of the diverging planes will represent the stresses along the bars to whose lines of resistance they are respectively normal.

It is obvious that the polyhedron of forces and the polyhedral frame are reciprocally related as follows: their numbers of edges are equal, and their corresponding pairs of edges perpendicular to each other; and the number of faces in each polyhedron is equal to the number of summits in the other.

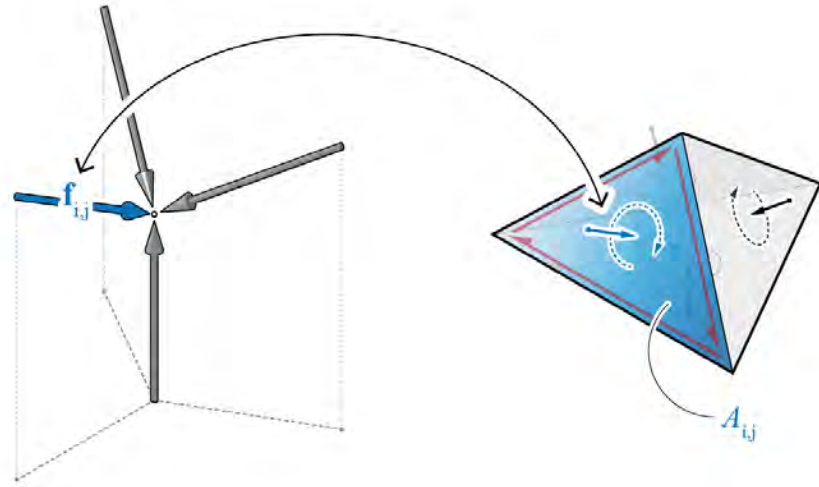
Glasgow, January 9, 1864.

**XVIII. On the Theory of the Velocity of Sound.**  
*By Professor CHALLIS, M.A., F.R.S., F.R.A.S.\**

**T**HE "Note" of Professor Tyndall "On the Velocity of Sound" in the Number of the Philosophical Magazine for last November, and the reference therein made to Dulong's experiments, have led me to see that the principles I have applied to this question admit of an extension which had not previously occurred to me. For this reason, and because I am desirous of making a few remarks on the notice taken of my researches by Prof. Le Conte in the article inserted in the January Number, I now revert to the subject.

The theoretical value of the velocity of sound which I have obtained, agreeing very closely with the observed value, is a purely mathematical deduction, on hydrodynamical principles, from the hypotheses that the medium is a perfect fluid, and that

\* Communicated by the Author.





## On the equilibrium of funicular polyhedral frames and convex polyhedral force diagrams

Masoud Akbarzadeh, Tom Van Mele, Philippe Block

ETH Zürich, Computational Technology in Architecture (CTA), Zürich, Switzerland

### HIGHLIGHTS

- Three-dimensional extension of graphic statics using polyhedral form and force diagrams.
- Defining the topological and geometrical relationships of 3D reciprocal diagrams.
- Design of tensegrity and tension only spatial structures with externally applied loads.
- Designing tensioned funicular spatial forms by aggregating convex force polyhedral cells.
- Characterization to maximize the geometry of the force and explore its effects on the form.

### ARTICLE INFO

Article history:  
Received 13 June 2014  
Accepted 15 January 2015

**Keywords:**  
Three-dimensional graphic statics  
Funicular diagrams  
Spatial geometry of force  
Convex force polyhedra  
Equilibrium graphs of architectural  
polyhedral frames

### ABSTRACT

This paper presents a three-dimensional extension of graphic statics using polyhedral form and force diagrams for the design of compression-only and tension-only spatial structures with externally applied loads. It explores the concept of 3D structural equilibrium based on Rastkin's original proposition for the equilibrium of spatial frames. It provides a definition for polyhedral reciprocal form and force diagrams, thus allowing self-equilibrating external forces and discusses their geometrical and topological characteristics. This paper furthermore provides a general procedure for constructing a pair of reciprocal polyhedral diagrams from a given polyhedron representing either the form or force diagram of a structural system. Using this method, this paper furthermore suggests a design strategy for finding unique funicular spatial forms in pure compression (or tension). Based on the construction of force diagrams through the aggregation of convex polyhedral cells finally, it discusses the effect of changes in the geometry of the force diagram on the geometry of the form diagram and the distribution of forces in it.

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### 1. Introduction

Graphic statics is a method for structural form finding that originates in the pre-digital era [1–4], but continues to be used and developed even today [5–7].

In graphic statics, the geometry and equilibrium of a structural system are represented by two reciprocal diagrams, the form and the force diagram. Since the geometrical relationship between these diagrams provides explicit control over both form and forces of a structure simultaneously, graphic statics is considered as an intuitive technique for structural design, relevant to architects, engineers, researchers, and students.

Despite its clear strengths and advantages, existing method of graphic statics has some limitations. The most important one is that a designer can only design three-dimensional structures by reducing them to a combination of two-dimensional sub-systems, for example, by using projections [3,4].

This paper therefore presents a three-dimensional version of graphic statics, using reciprocal polyhedral form and force diagrams for the design and analysis of spatial frames with externally applied loads.

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#### 1.1. Previous work

Reciprocal diagrams are the basis of the conventional graphic methods of structural form finding. This section provides a brief summary of the use of reciprocal diagrams and their geometrical relationship in structural design.

#### 2D reciprocal diagrams

The concept of investigating static equilibrium through geometric constructions involving polygons of forces first appeared

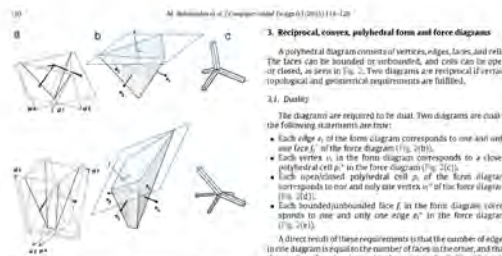


Fig. 6. (a) The polyhedral frame with its reciprocal polyhedral force diagram. (b) The reciprocal of a given polyhedron in force diagram form. (c) A funicular form in pure compression.

(Fig. 6) depicts the application of Rastkin's theory to two polyhedral frames, consisting of three and four bars respectively (Fig. 7). (a) shows the planes that are perpendicular to the bars of the frame, and diverge from a point (Fig. 6b). In both cases, the planes form an open polyhedron. A line perpendicular to an additional force (for example, an applied load) (shown the polyhedron) (Fig. 7c) leads to the force equilibrium for the whole. The area of the faces of the closed polyhedron are proportional to the magnitudes of the forces in the corresponding bars of the frame. Each choice of closing face thus results in a different distribution of forces in the frame (Fig. 7c).

In the knowledge of the authors, Rastkin never proved his theory. Therefore, we provide here a short proof based on the divergence theorem [25,26].

Let  $P$  be a closed polyhedron in  $n$ -D with enclosing faces  $f$ , and volume  $V$ , and  $\mathbf{a}$  an arbitrary, constant vector field in  $n$ . Each face, furthermore, has area  $A$  and normal vector  $\mathbf{n}$ . According to the divergence theorem, the total outward flux  $\Phi$  through the closed surface of the polyhedron is equal to the divergence  $\nabla \cdot \mathbf{a}$  over the enclosed region, which is two times  $V$  in 3D:

$$\Phi = \oint_P \mathbf{a} \cdot \mathbf{n} dA = \int_V \nabla \cdot \mathbf{a} dV = 2V \nabla \cdot \mathbf{a}$$

For the polyhedron, we can then write:

$$\sum_f \mathbf{a} \cdot \mathbf{n} A_f = 2V \nabla \cdot \mathbf{a}$$

"We view the sum of all area-weighted vectors  $\sum_f \mathbf{a} \cdot \mathbf{n} A_f$  of the polyhedron here as 'area' with a unit vector  $\mathbf{n}$ . If the forces  $\mathbf{F}$  applied to a point in space are perpendicular to the faces of a polyhedron, and their magnitudes proportional to the area of the faces, the sum of these forces must be zero, leaving the point in equilibrium:

$$\sum_f \mathbf{F}_f = \sum_f \mathbf{a} \cdot \mathbf{n} A_f = 0$$

### 3. Reciprocal, convex, polyhedral form and force diagrams

A polyhedral diagram consists of vertices, edges, faces, and cells. The faces can be bounded or unbounded, and cells can be open or closed, as seen in Fig. 8. Two diagrams are reciprocal if vertex, topological and geometrical requirements are fulfilled.

#### 3.1. Duality

The diagrams are required to be dual. Two diagrams are dual if the following statements are true:

- Each edge  $e$  of the form diagram corresponds to one and only one face  $f$  of the force diagram (Fig. 8b).
- Each vertex  $v_i$  in the form diagram corresponds to a closed polyhedral cell  $p_i$  in the force diagram (Fig. 8c).
- Each unbounded polyhedral cell  $p_i$  of the force diagram corresponds to one and only one vertex  $v_i$  of the form diagram (Fig. 8d).
- Each bounded/unbounded face  $f$  in the form diagram corresponds to one and only one edge  $e$  in the force diagram (Fig. 8e).

A direct result of these requirements is that the number of edges in one diagram is equal to the number of faces in the other, and that the number of vertices is equal to the number of cells (Fig. 8b–e).

(Fig. 8a, b) illustrates these topological relationships. The elements of the form diagram are labeled with lower case letters, and the elements of the force diagram (tagged with an asterisk) (\*)

#### 3.2. Planarity and perpendicularity

If, in addition, all faces are planar, and all edges perpendicular to their dual faces, the diagrams are reciprocal. The force diagram then represents the structural equilibrium of the system of forces represented by the form diagram, with the force in each edge of the form diagram  $\mathbf{F}_f$  equal to the area of its dual face  $A_f$ .

Note that due to the presence of external loads and reaction forces, the form diagram has both bounded and unbounded faces and open and closed cells. The force diagram, on the other hand, has only bounded faces and closed cells. The inside faces of the form diagram correspond to the external forces. All other faces represent the internal forces of the form.

The closed cell formed by the inside faces represents global equilibrium of all external forces. Each internal cell  $p_i$  represents the equilibrium of its dual vertex  $v_i$ .

(Fig. 8c, d) illustrates the perpendicular relationships between form and force diagrams in 3D. The elements of the reciprocal force diagram fulfill with an asterisk (\*)

#### 3.3. Geometry

In this paper we focus on the equilibrium of compressive or tension-only structures. A system of forces in a pure compression (or tension), if its force polyhedron is a proper cell decomposition of space, is a tensegrity structure. This requirement is analogous to the conditions for equilibrium of spatial wires described by Ash et al. [27]. A force polyhedron is a proper cell decomposition of space, if the following statements are true:

- every point in space belongs at least to one cell;
- the cells have disjoint interiors;
- the decomposition is face to face (i.e. every face of one cell is a complete face of another cell).

#### 4. Constructing reciprocal polyhedrons

In this section, we describe a straightforward algorithm for constructing a pair of reciprocal diagrams from a given polyhedron representing either the form or force diagram of a structural

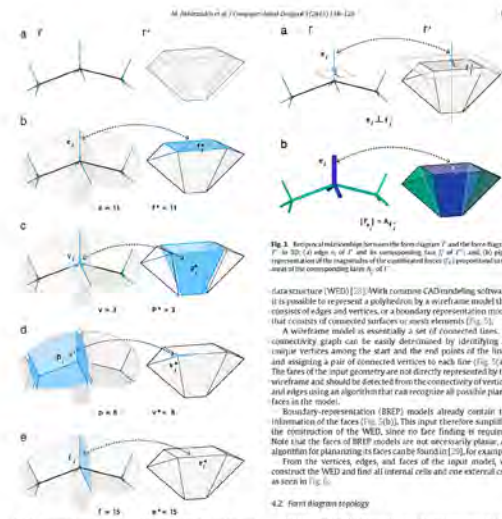


Fig. 7. Reciprocal relationship between form diagram  $F$  and the force diagram  $F^*$  in 3D. (a)  $F$  and  $F^*$  in 3D. (b)  $F$  in 3D. (c)  $F$  in 2D. (d)  $F$  in 2D. (e)  $F$  in 2D.

system. For simplicity of the explanation we assume here the given diagram is the force diagram.

As depicted in Fig. 9, the algorithm consists of three main sections: (1) constructing the force diagram from a given geometric representation and extracting its topology (see Section 4.2); (2) generating the topology of the form diagram (see Section 4.3); and (3) imposing perpendicularity (see Section 4.4).

#### 4.1. Force diagram topology

The first step is to determine the topology of the force diagram. The topology of a polyhedron can be described with a signed edge

data structure (WED) [28]. With a computer (CAD/rendering software), it is possible to represent a polyhedron by a vertex-edge model that consists of edges and vertices, or a bounding representation model that consists of connected surfaces or mesh elements (Fig. 9).

A vertex-edge model is essentially a set of connected lines. Its connectivity graph can be easily determined by identifying all unique vertices among the start and the end points of the lines, and assigning a pair of connected vertices to each line (Fig. 9a). The faces of the input geometry are not directly represented by the vertex-edge and should be detected from the connectivity of vertices and edges using an algorithm that can recognize all possible planar faces in the model.

Boundary representation (BREP) models already contain the information of the faces (Fig. 9b). This input therefore simplifies the reconstruction of the WED, since no face finding is required. Note that the faces of BREP models are not necessarily planar. An algorithm for planarizing in faces can be found in [29]. For example, from the vertices, edges, and faces of the input model, we construct the WED and find all internal cells and outer external cells, as seen in Fig. 9c.

#### 4.2. Force diagram topology

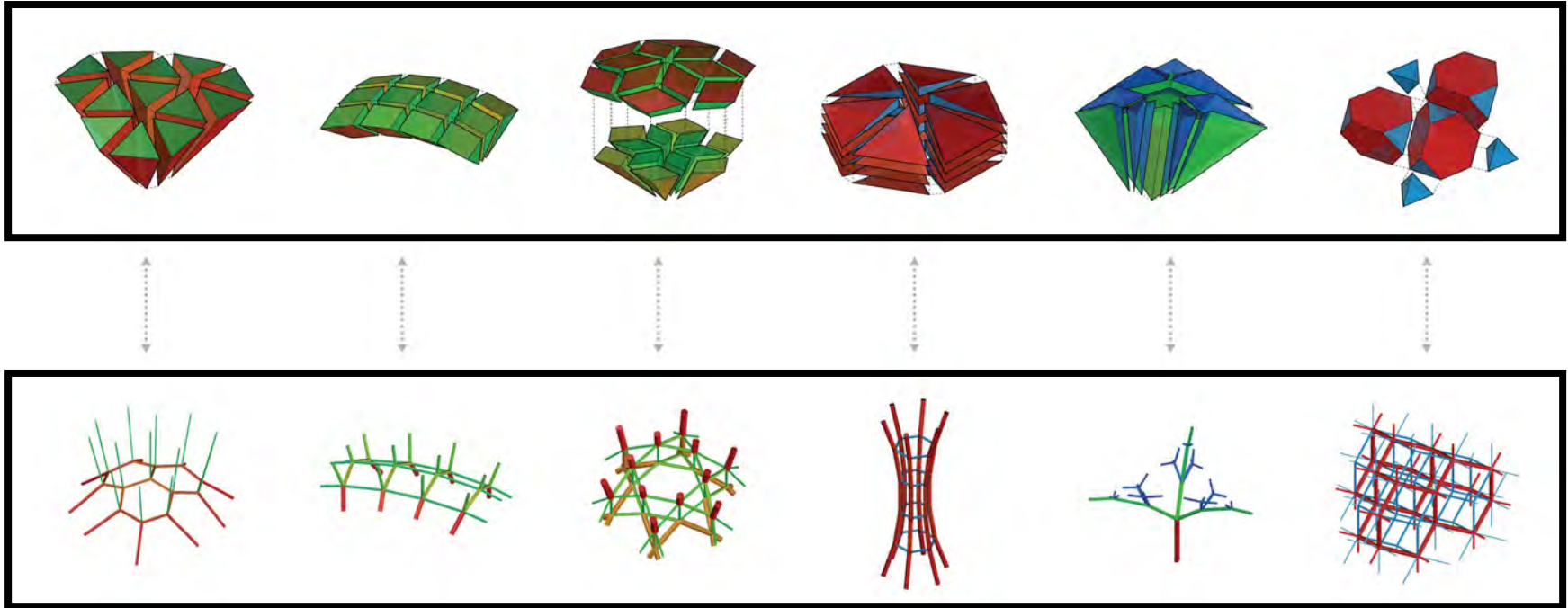
The connectivity of the force diagram follows immediately from the adjacency graph of the polyhedral cells of the force diagram. We can, therefore, construct a topologically correct form diagram by connecting the centroids of adjacent cells,  $v_i$ ,  $v_j$ , in the force diagram (Fig. 9a). Each internal cell adjacent to the external cell is furthermore connected to the centroids of all external faces (Fig. 9b).

#### 4.3. Form diagram perpendicularity

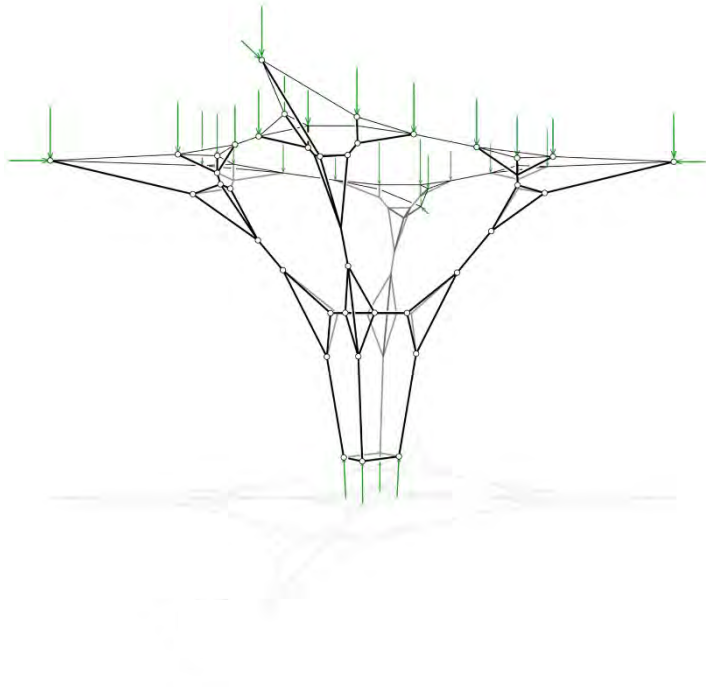
So far, we constructed a polyhedral frame that has the topology of the desired form diagram. However, the edges of this polyhedral frame are generally not perpendicular to the faces of the force diagram.

We impose perpendicularity through an iterative procedure in which all iteration consists of vertices. A similar algorithm based on the face for all force diagrams described by Kijewski et al. [30].

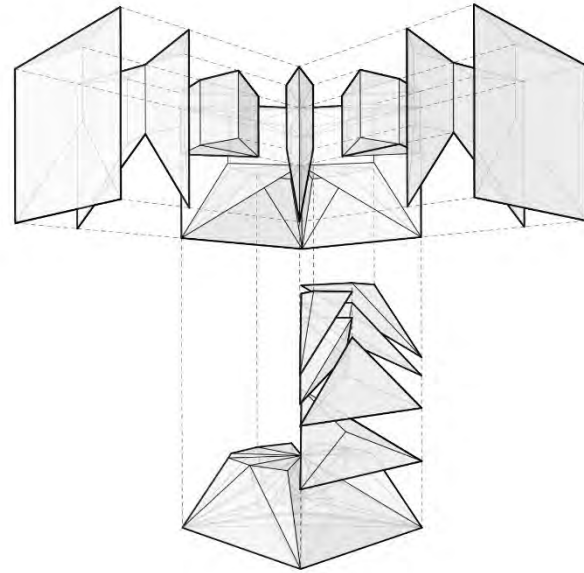
### Polyhedral force diagram



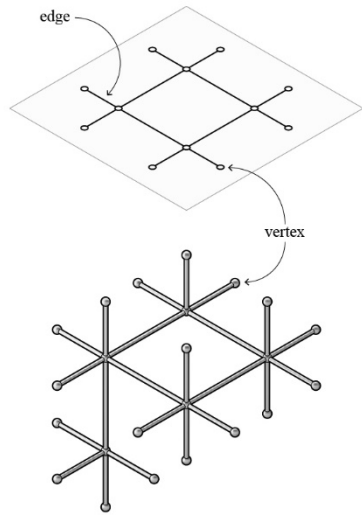
### Polyhedral form diagram



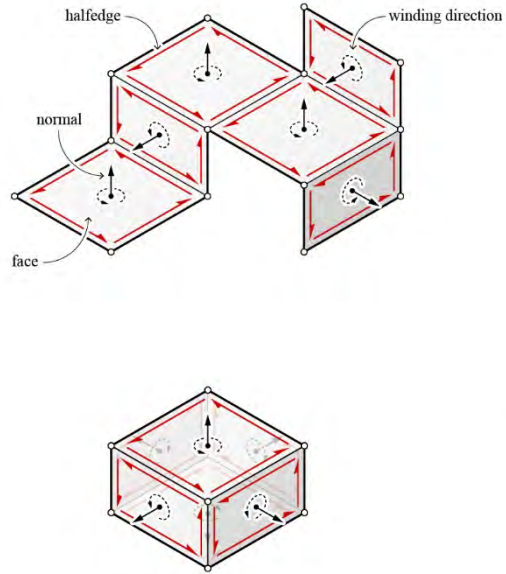
**Polyhedral Form Diagram**



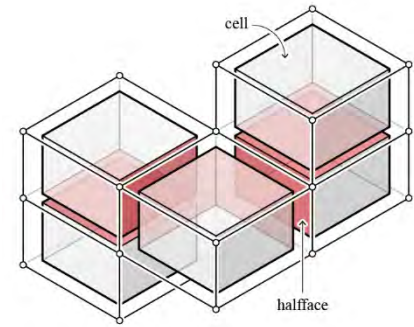
**Polyhedral Force Diagram**



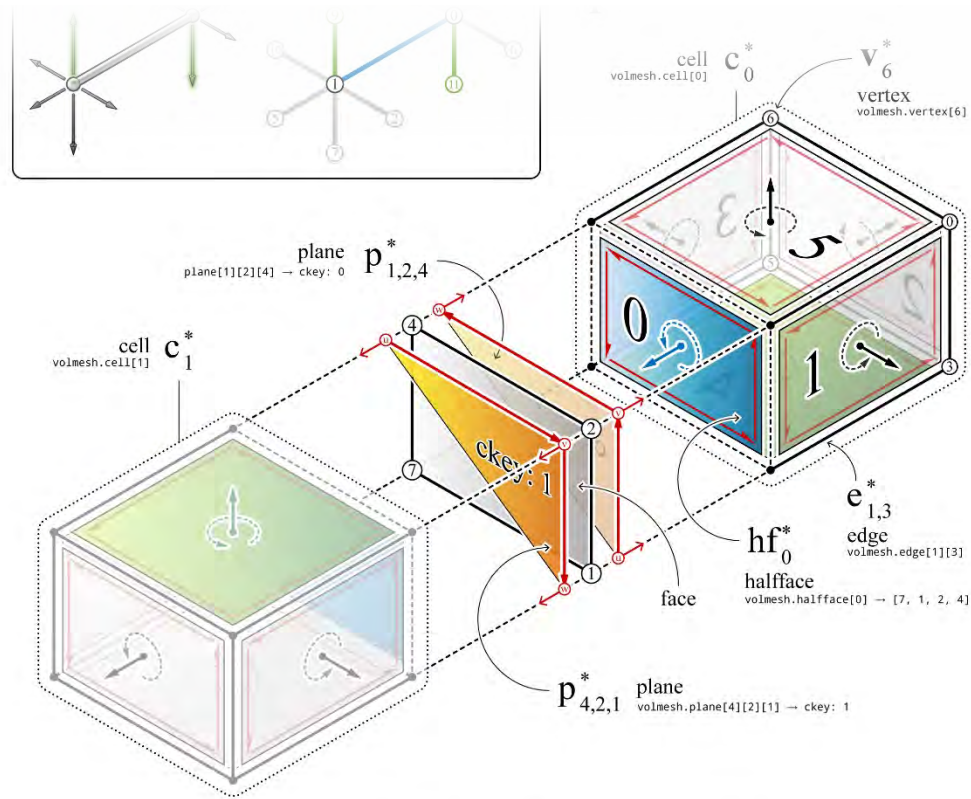
**network**  
network of vertices



**mesh**  
network of faces



**volmesh**  
network of cells



vertex

| vkey | attr                                   |
|------|----------------------------------------|
| 0    | { 'x': 10.0<br>'y': 10.0<br>'z': 7.0 } |
| 1    | { 'x': 10.0<br>'y': 0.0<br>'z': 0.0 }  |

edge

| u | v | attr |
|---|---|------|
| 0 | 2 | { }  |
| 1 | 2 | { }  |
|   | 3 | { }  |
| 2 | 4 | { }  |
| 3 | 0 | { }  |
|   | 5 | { }  |

halfface

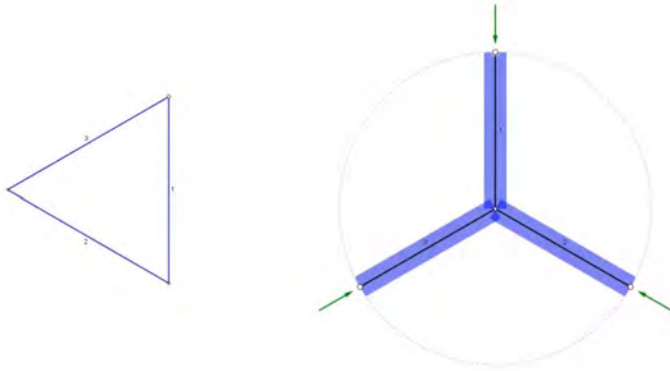
| hkey | vkeys |
|------|-------|
| 0    | 7     |
|      | 1     |
|      | 2     |
|      | 4     |
| 1    | 0     |
|      | 2     |

cell

| ckey | u | v | hkey |
|------|---|---|------|
| 0    | 0 | 2 | 1    |
|      |   | 3 | 2    |
|      |   | 6 | 5    |
| 1    | 7 | 4 |      |
|      |   | 2 | 0    |
|      |   | 3 | 1    |

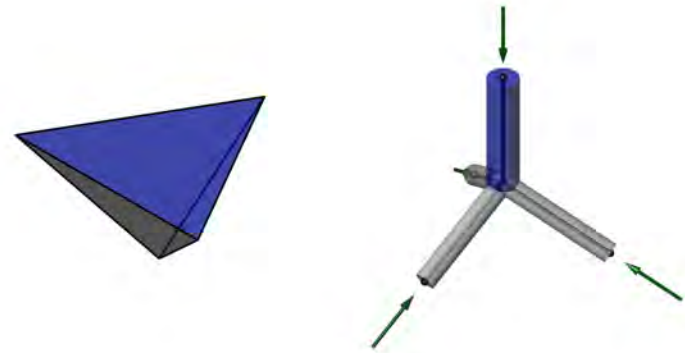
plane

| u | v | w | ckey | hkey |
|---|---|---|------|------|
| 0 | 2 | 1 | 0    | 4    |
|   | 3 | 1 | 1    | 5    |
|   | 6 | 4 | 0    | 5    |
| 1 | 7 | 4 | 1    | 5    |
|   | 0 | 1 | 4    | 0    |
|   | 3 | 0 | 0    | 5    |



## 2D graphic statics

Control lengths of force polygons → Control force magnitudes



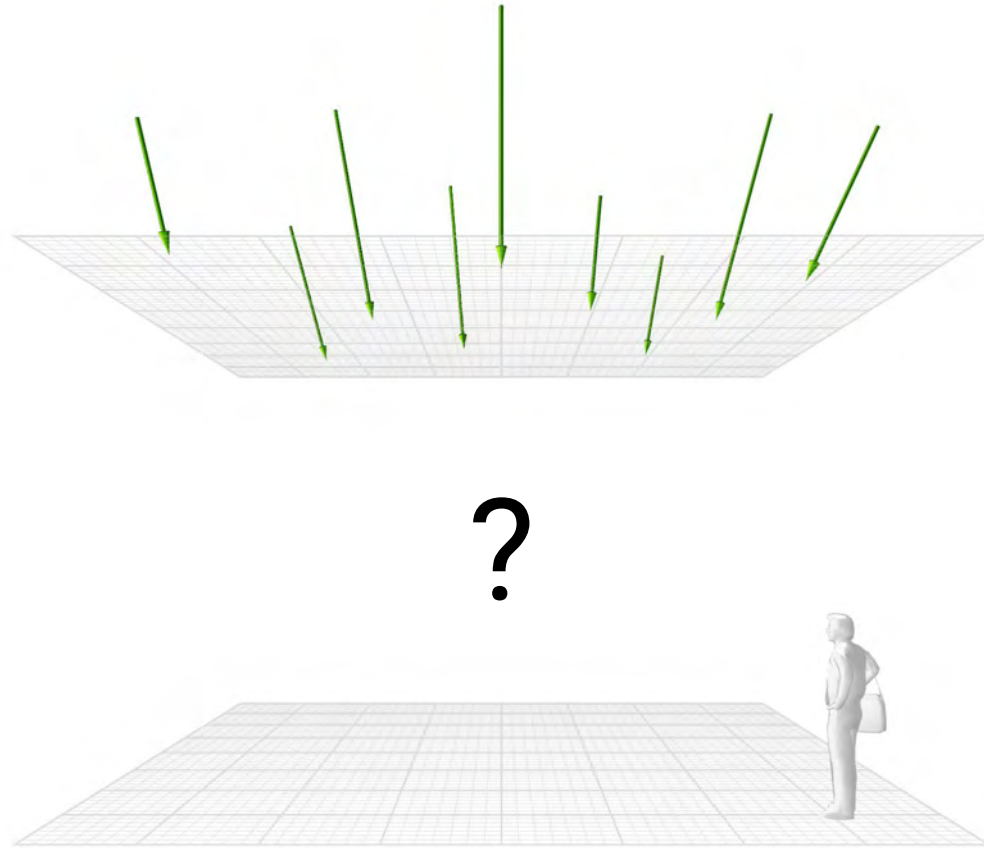
## 3D graphic statics

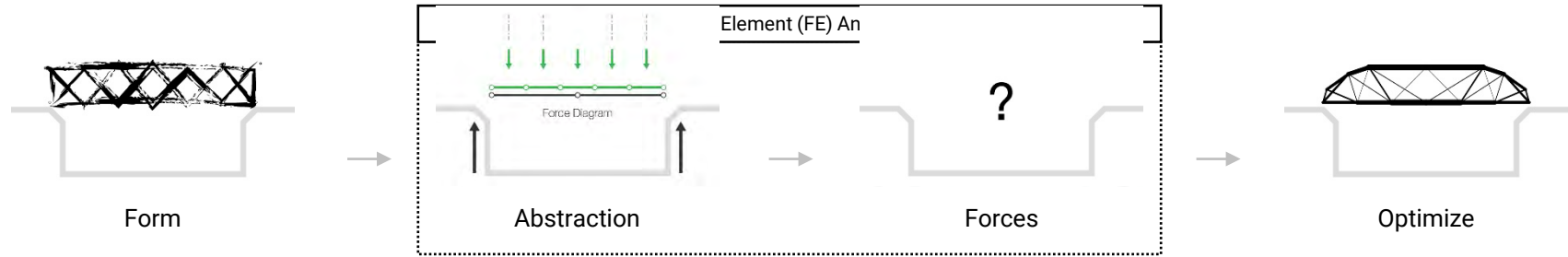
Control areas of force polyhedra → Control force magnitudes

↑  
NOT as straight forward or intuitive

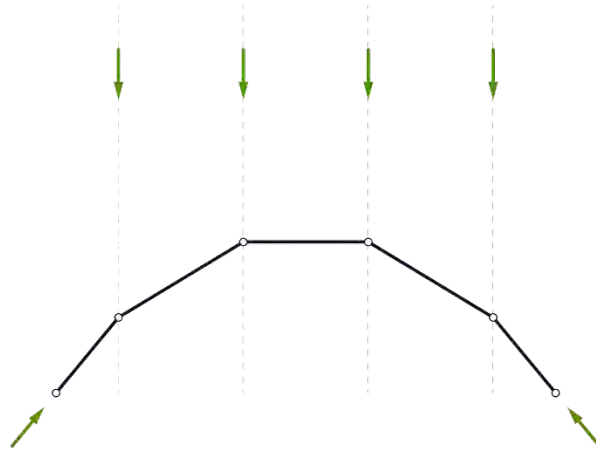




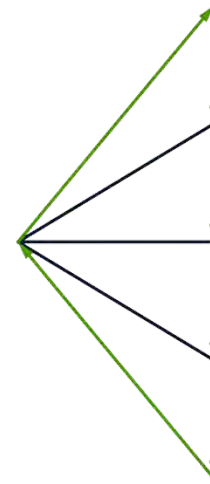




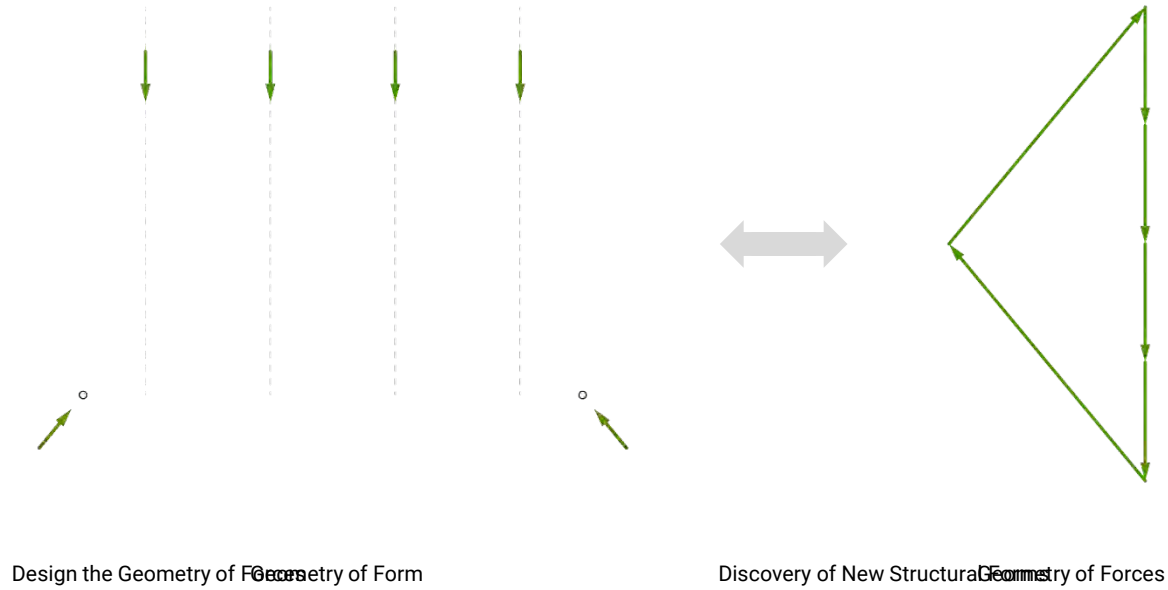
## Proposed Force-driven Design

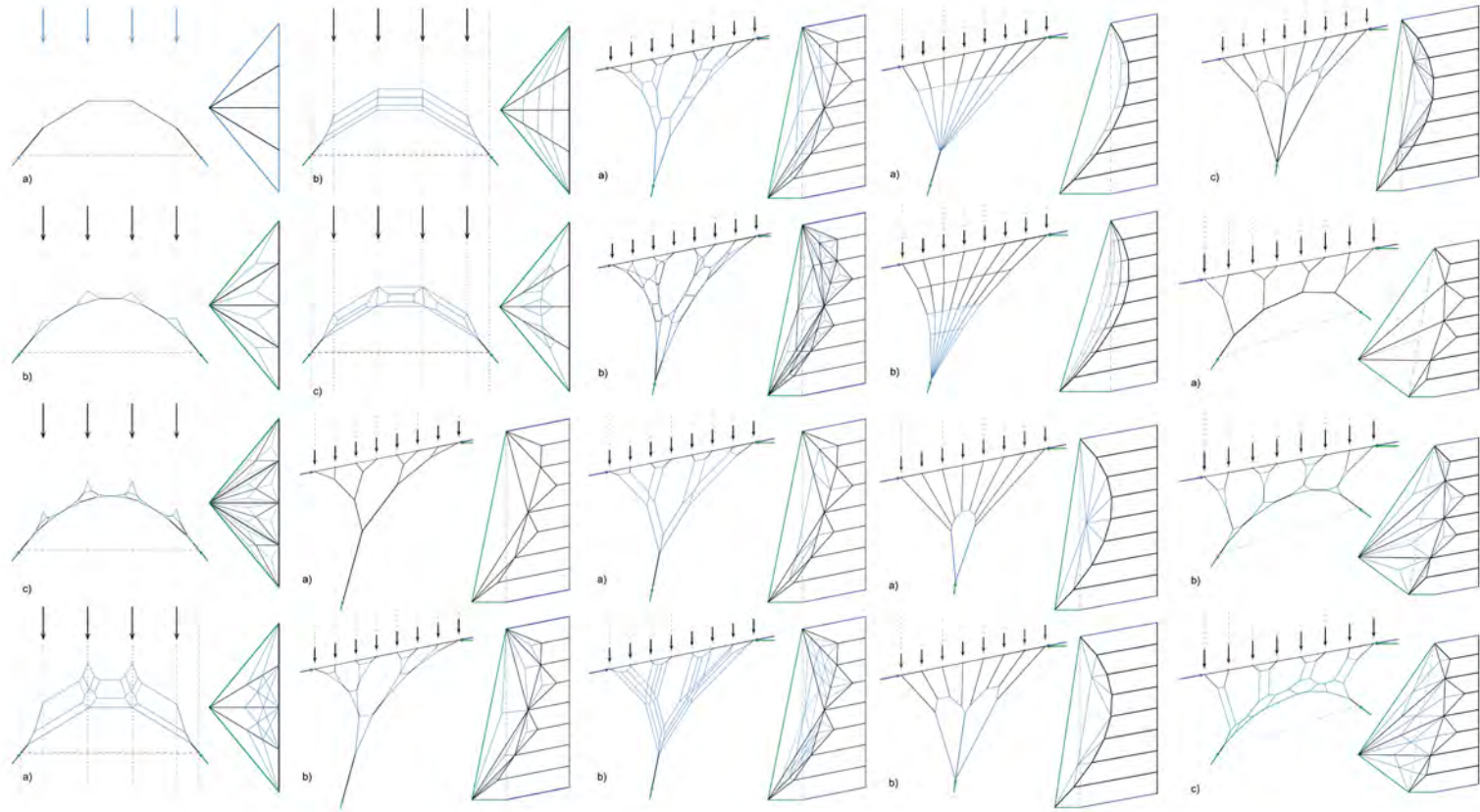


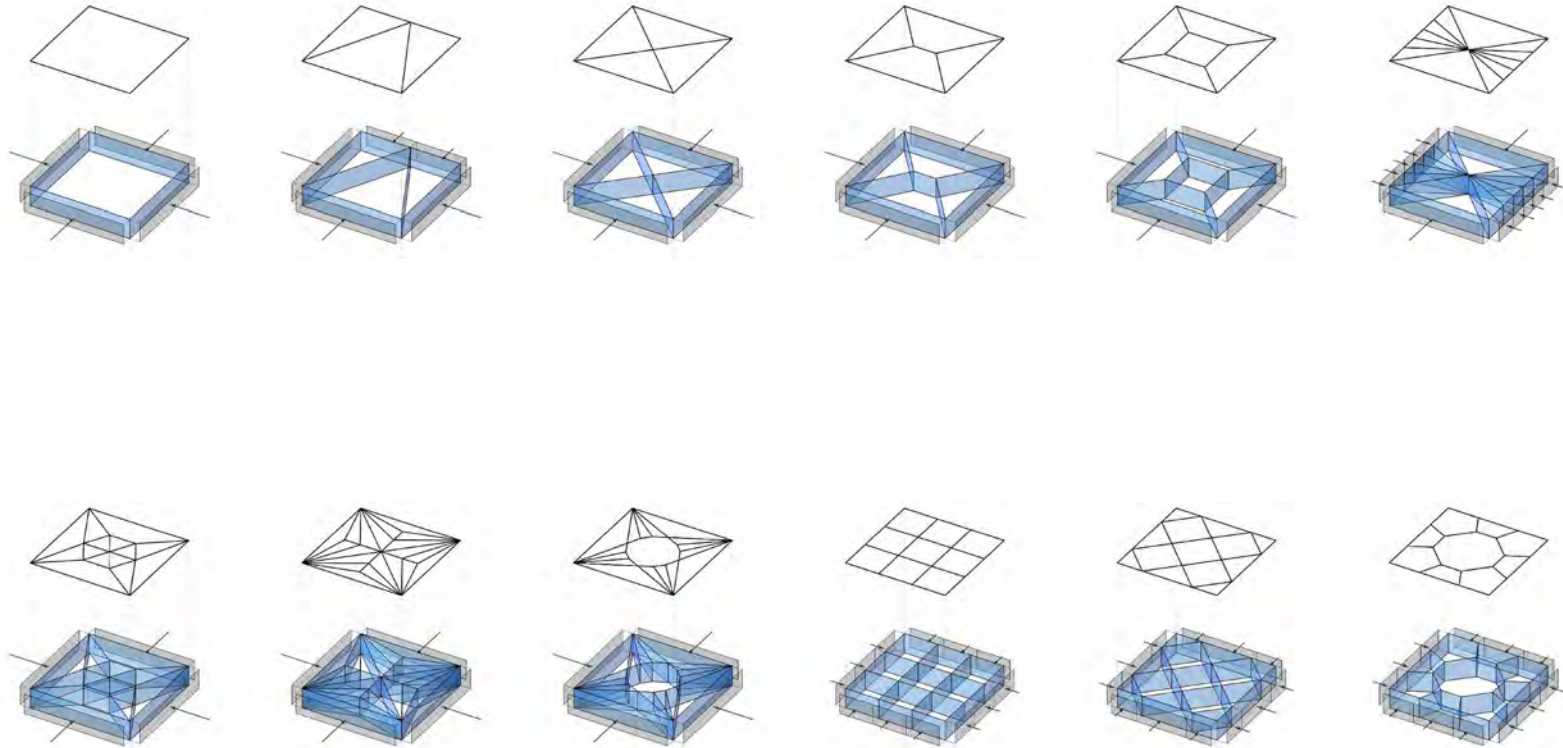
Geometry of Form

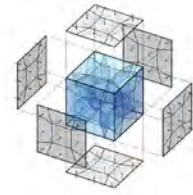
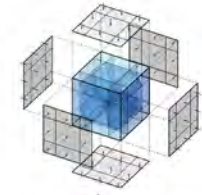
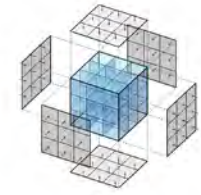
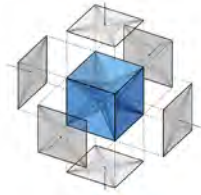
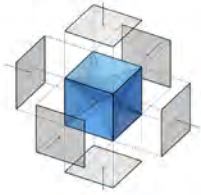
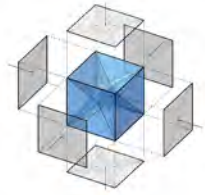
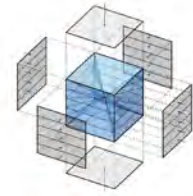
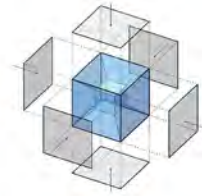
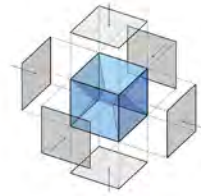
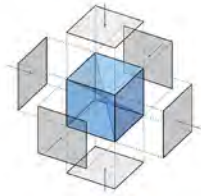
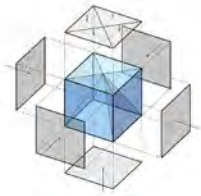
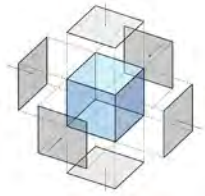


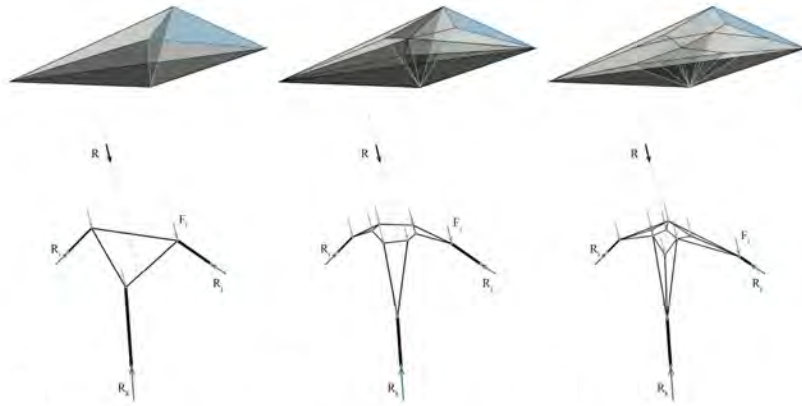
Geometry of Forces



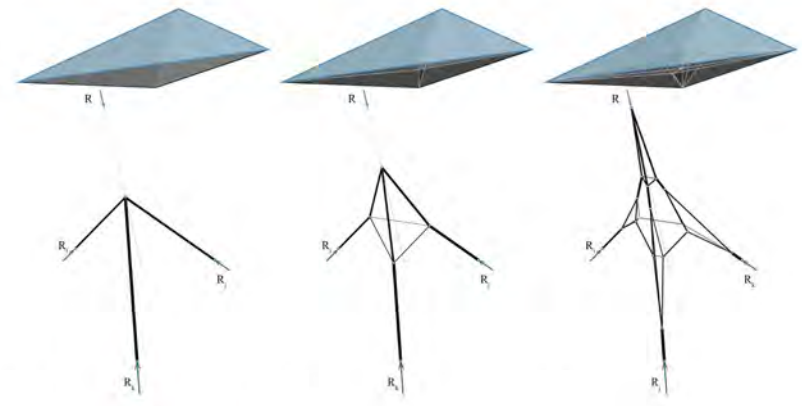




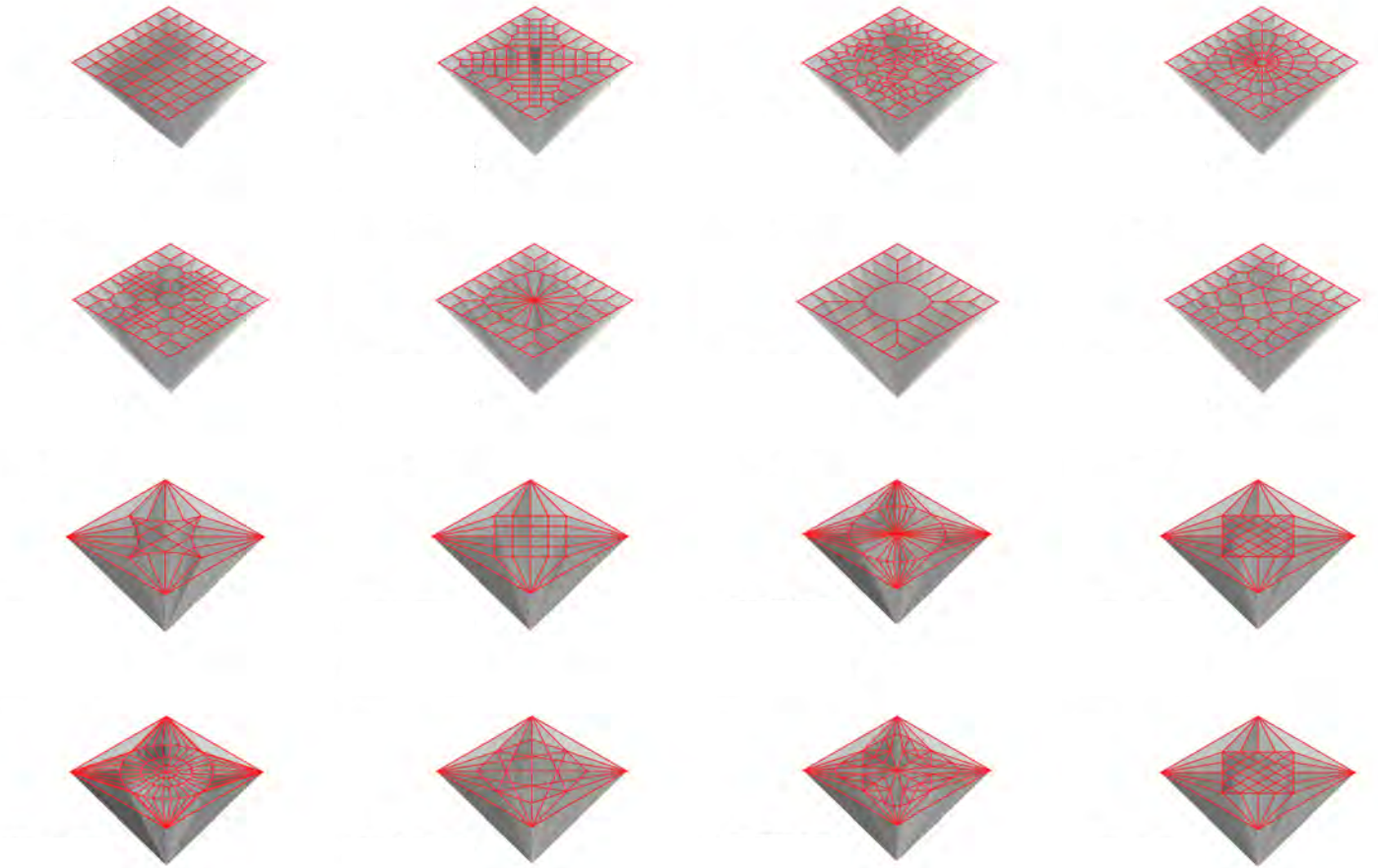


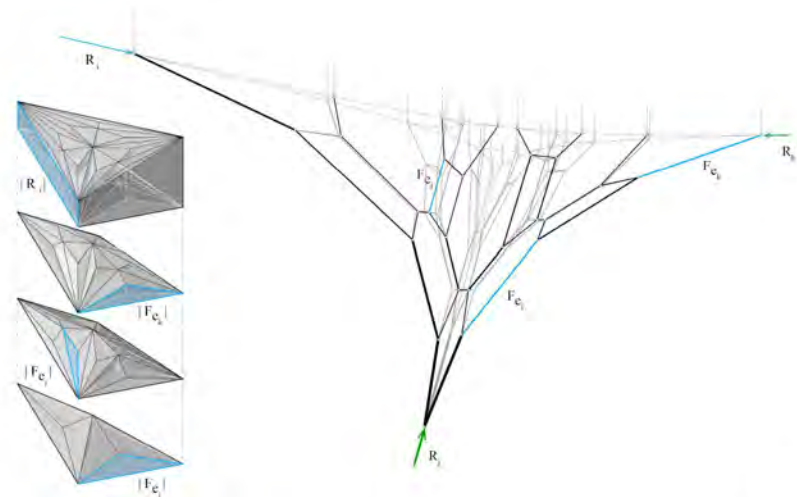
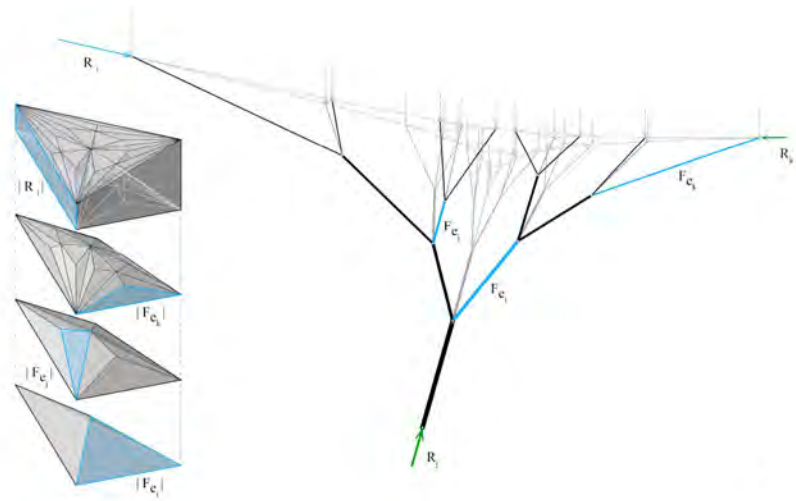


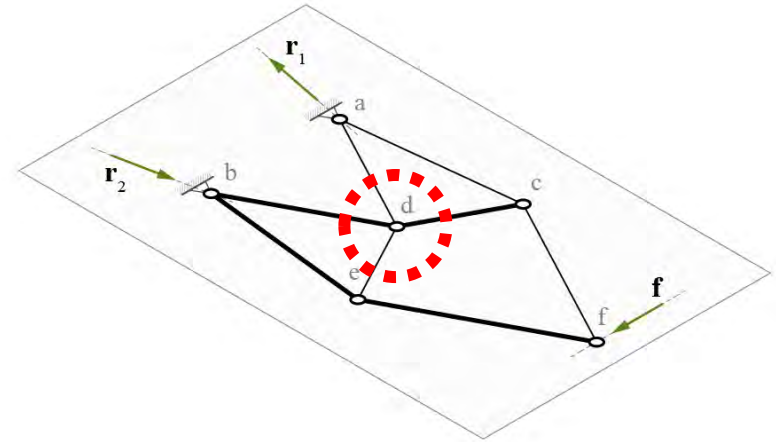
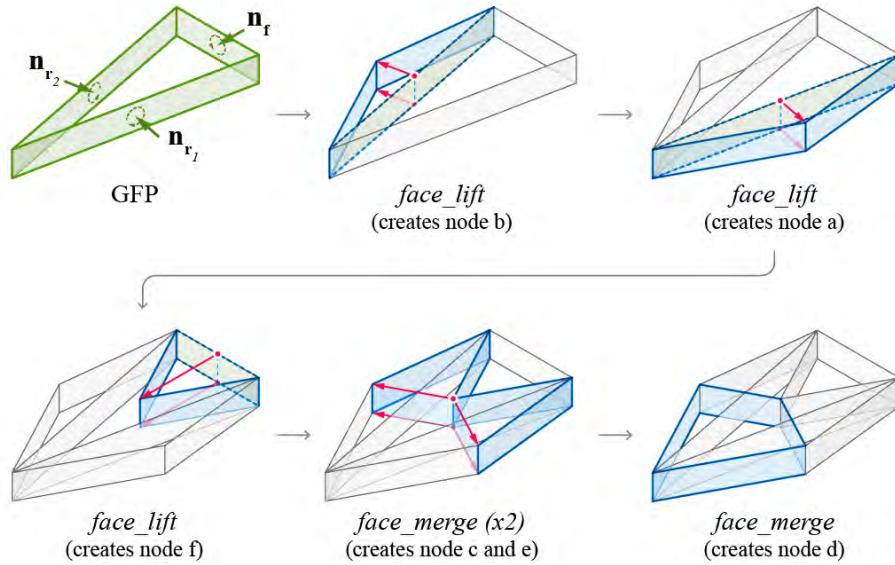
Subdivision of outer faces → changes boundary conditions

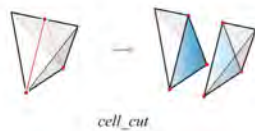
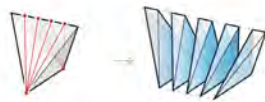
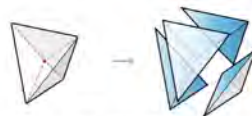


Subdivision of cells → refinement of predefined form

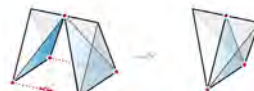
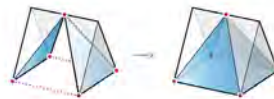






*cell\_cut**cell\_cut\_radial**cell\_cut\_face**cell\_subdivide*

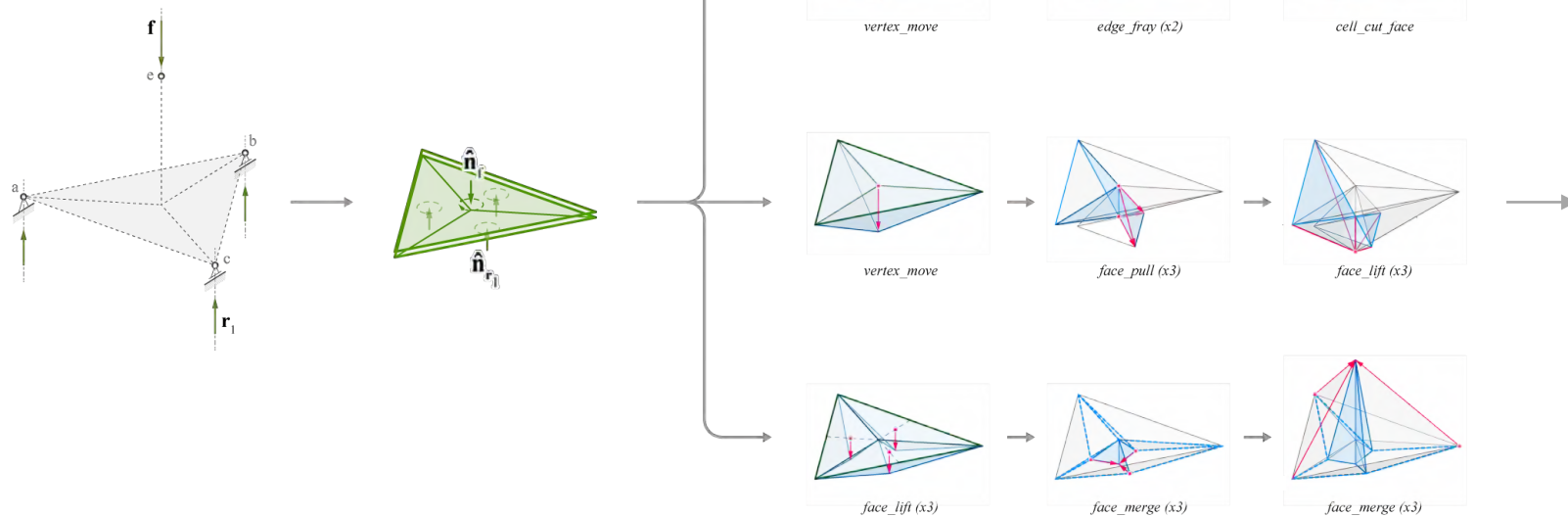
Subdivision (decomposition)

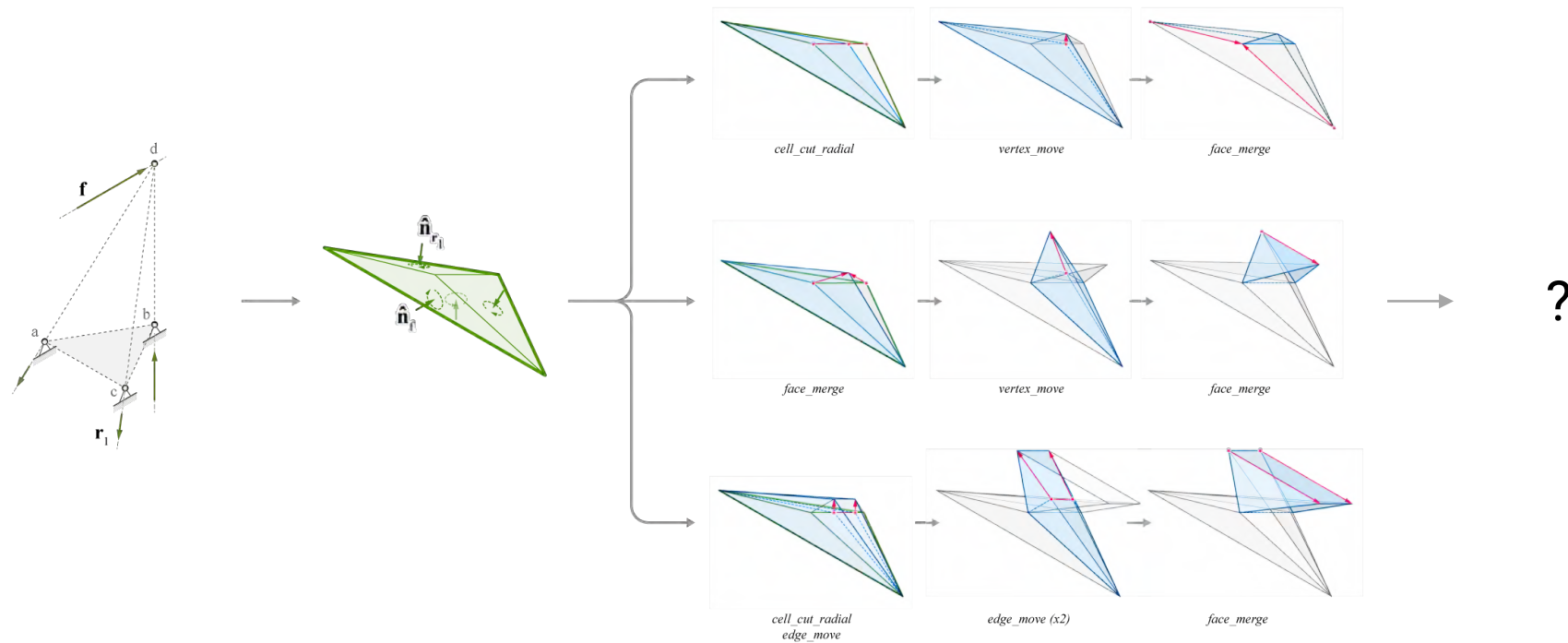
*edge\_split**face\_lift**face\_merge**face\_fill*

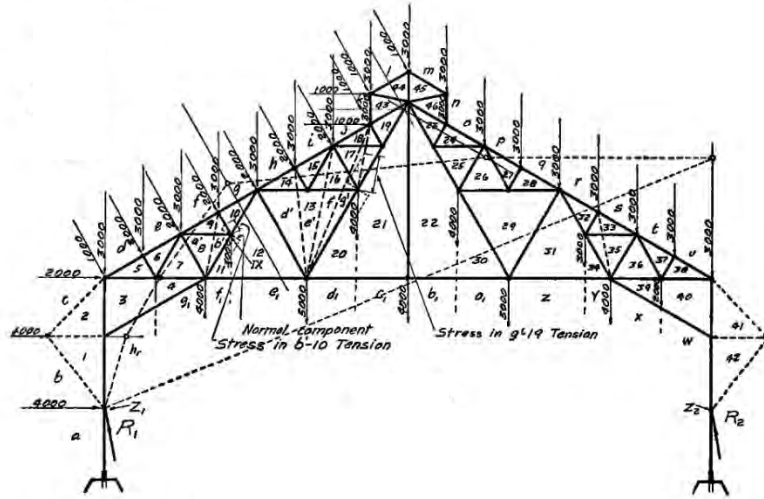
Transformation (additive)

*vertex\_move**edge\_move**face\_move**face\_push*

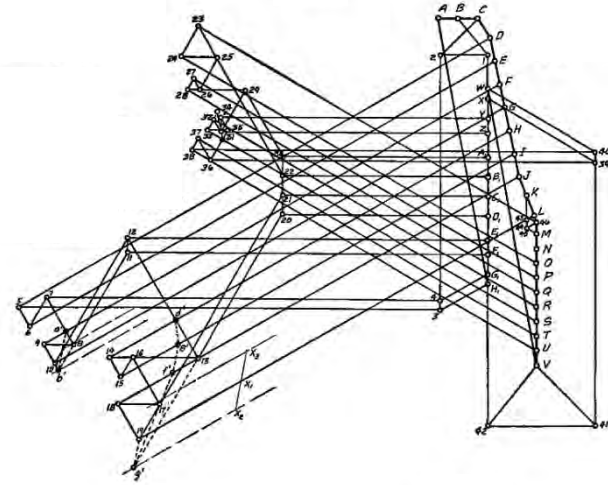
Modification



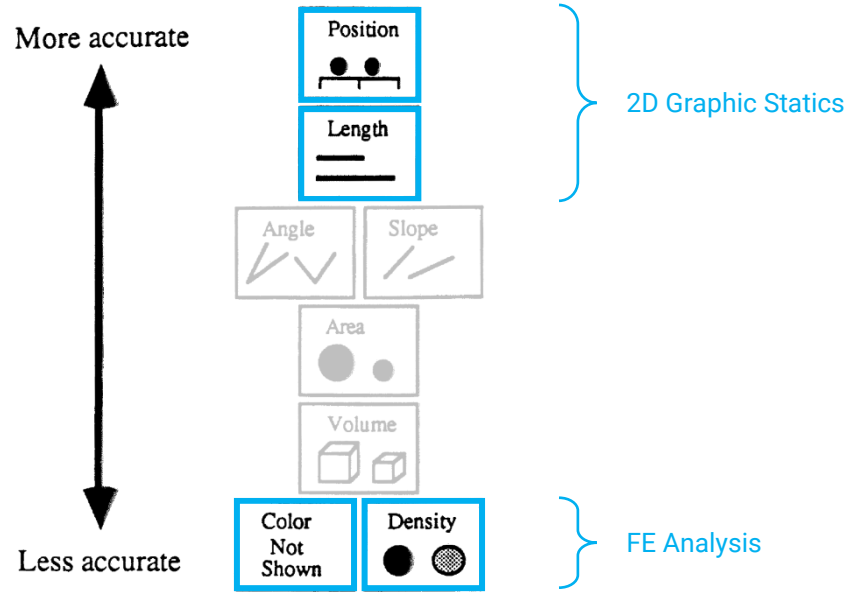




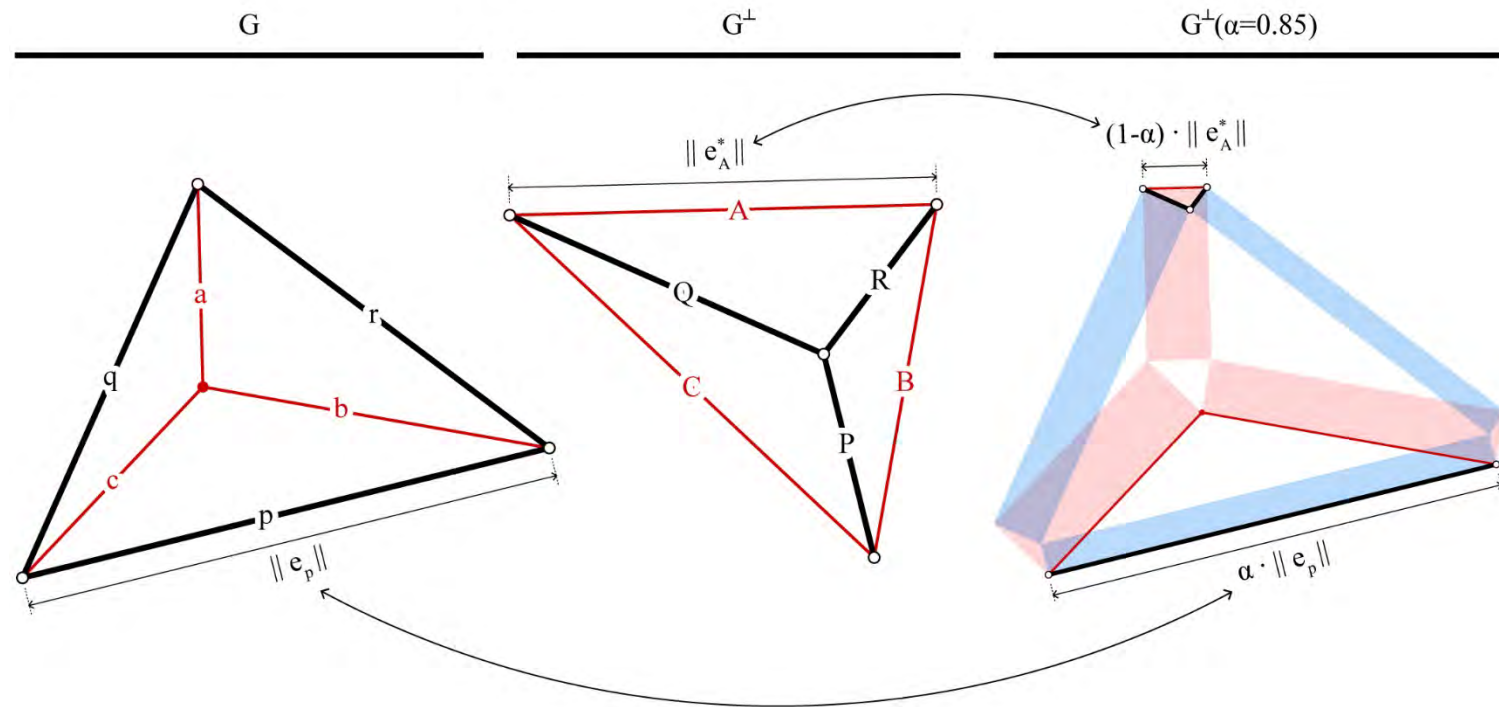
Form Diagram



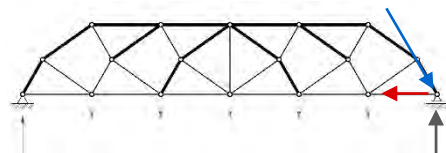
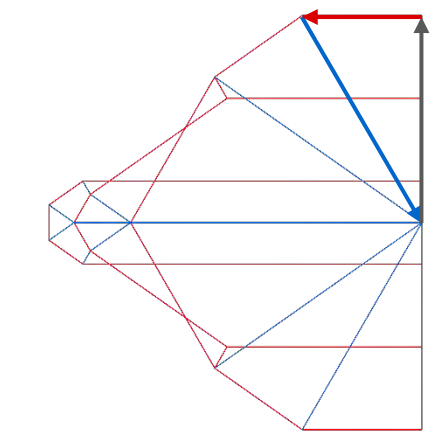
Force Diagram



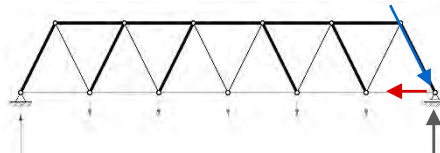
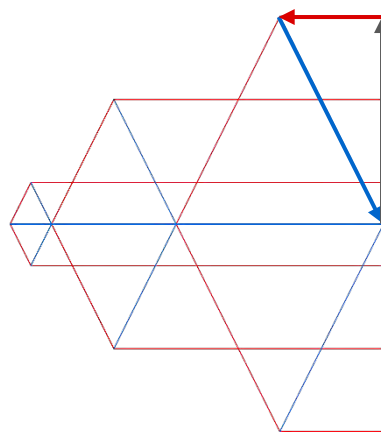
### Accuracy and Effectiveness of Human Retinal Perception



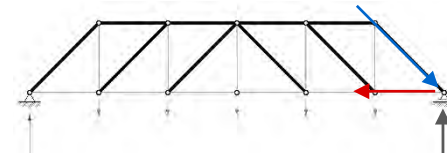
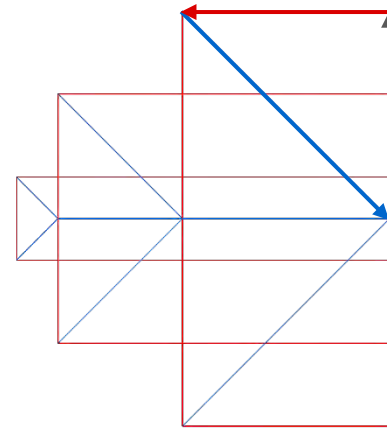
# "Maxwell" Diagram (parallel orientation)



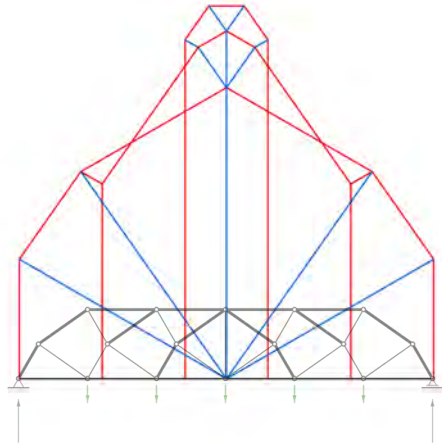
Michell



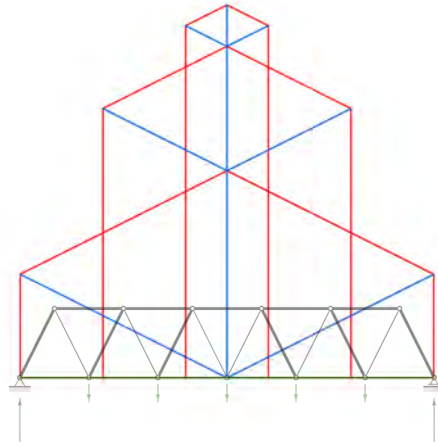
Warren



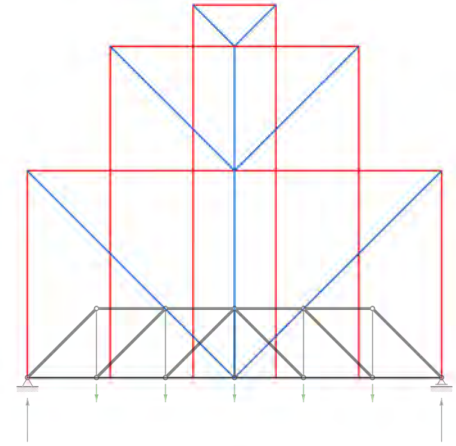
Howe



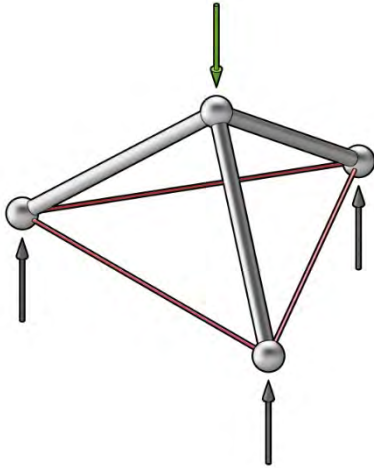
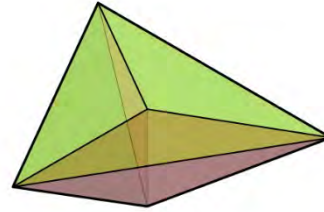
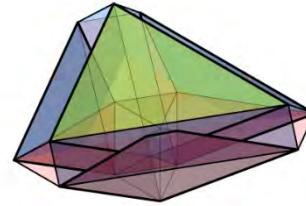
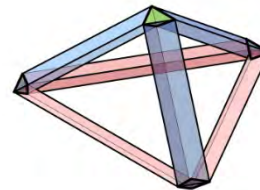
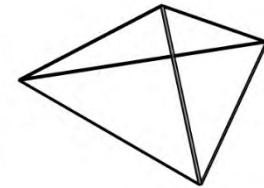
Michell

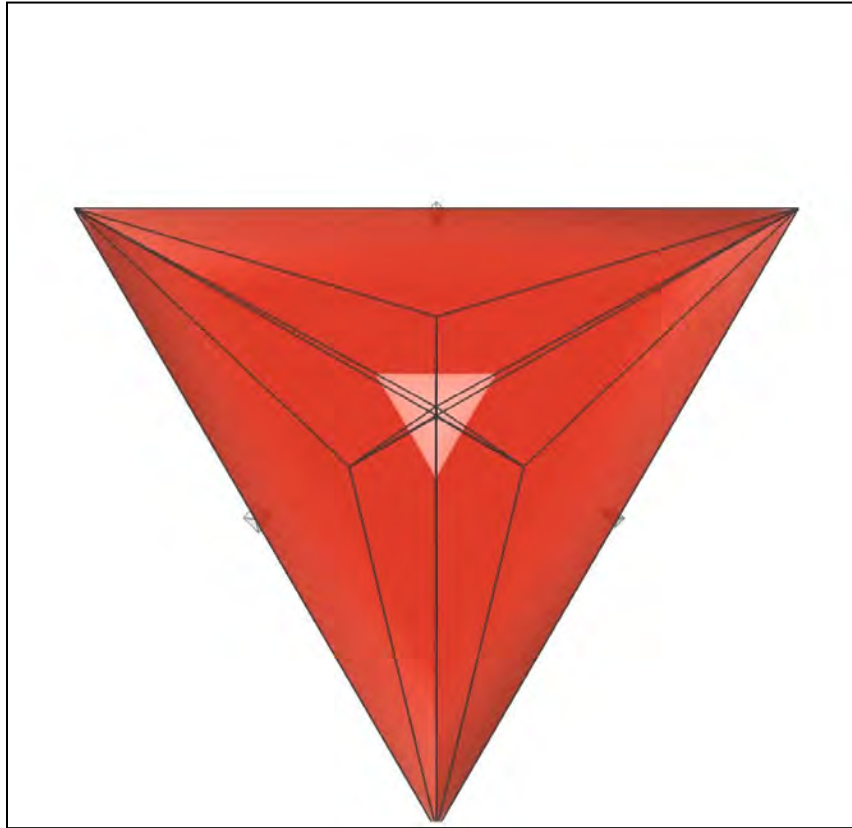


Warren



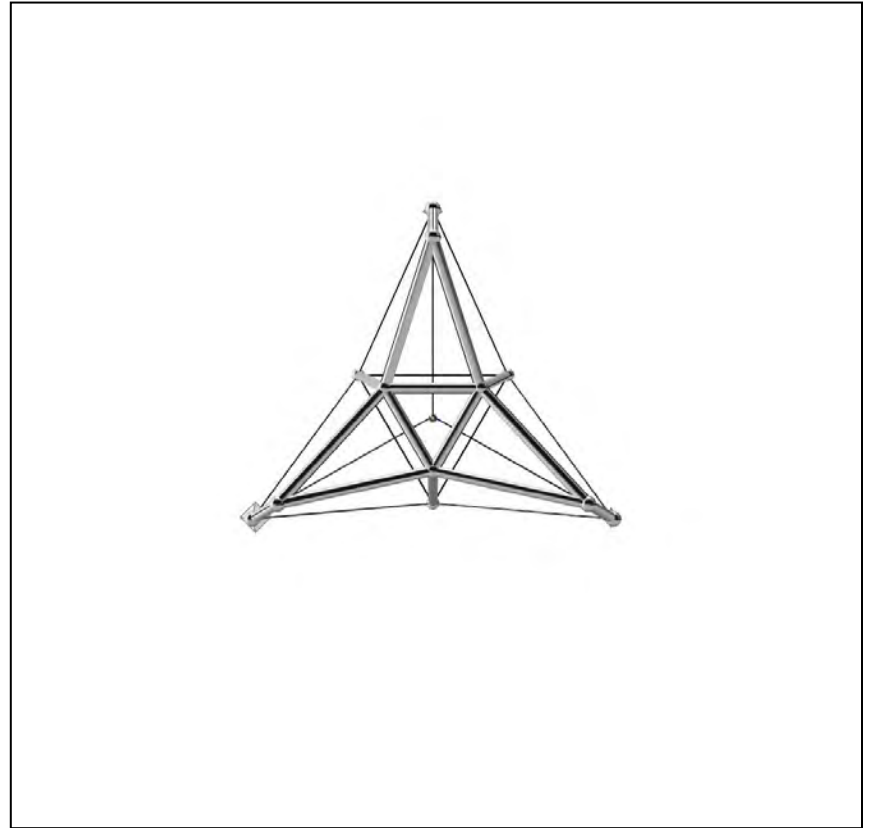
Howe

a)  $\Gamma$ b)  $\Gamma^\perp(\alpha)$  $\alpha = 0$  $\alpha = 0.25$  $\alpha = 0.5$  $\alpha = 0.75$  $\alpha = 0.90$  $\alpha = 0.99$



Interactive Polyhedral Force Diagram

← Synchronized Interface →



Form Diagram

2017 Seoul Biennale of Architecture and Urbanism

# MycoTree

## Beyond Mining - Urban Growth

### Sustainable Construction KIT Karlsruhe

Karsten Schlesier  
Felix Heisel  
Dirk Hebel

### Block Research Group ETH Zürich

Juney Lee  
Matthias Rippmann  
Tomás Méndez Echenagucia  
Andrew Liew  
Noelle Paulson  
Tom Van Mele  
Philippe Block

### Future Cities Laboratory Singapore-ETH

Nazanin Saeidi  
Alireza Javadian  
Adi Reza Nugroho  
Robbi Zidna Ilman  
Erlambang Adjidarma  
Ronaldiaz Hartantyo  
Hokie Christian  
Orion Tan  
Sheng Yu  
Kelly Cooper

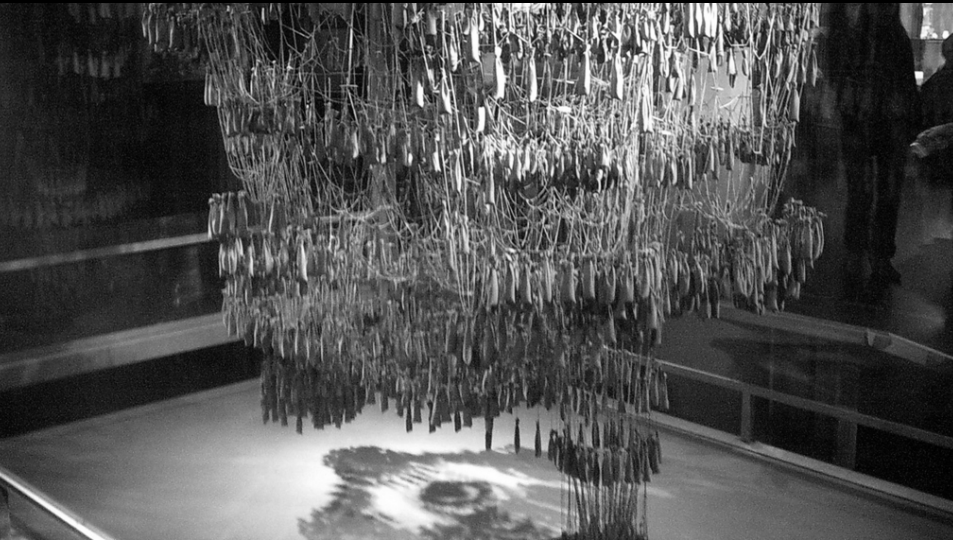


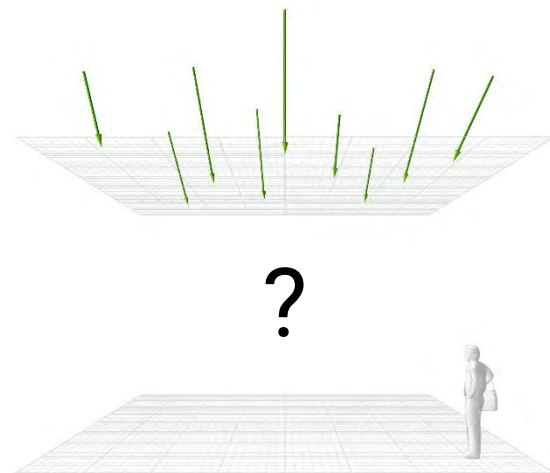


이 작품은 2011년 제1회 서울국제미술제에서 선보인 작품으로, 전통적인 한국 미술의 형태와 색채를 현대적으로 재해석한 작품이다. 작품은 주로 노란색과 흰색을 사용하며, 다양한 크기와 형태의 블록을 조합하여 구성되었다. 작품의 형태는 전통적인 한국 미술의 형태를 모티브로 하여, 현대적인 기법으로 재구성된 형태를 띠고 있다. 작품은 공간 속에서 빛을 받아 더욱 생동감 있게 표현되며, 관람객의 시각을 사로잡는 효과를 얻고 있다.

작품명: **노란색의 기억**  
작가: **김민준**  
소재: **노란색 블록, 흰색 블록**  
크기: **200cm x 150cm x 100cm**  
제작년도: **2011년**







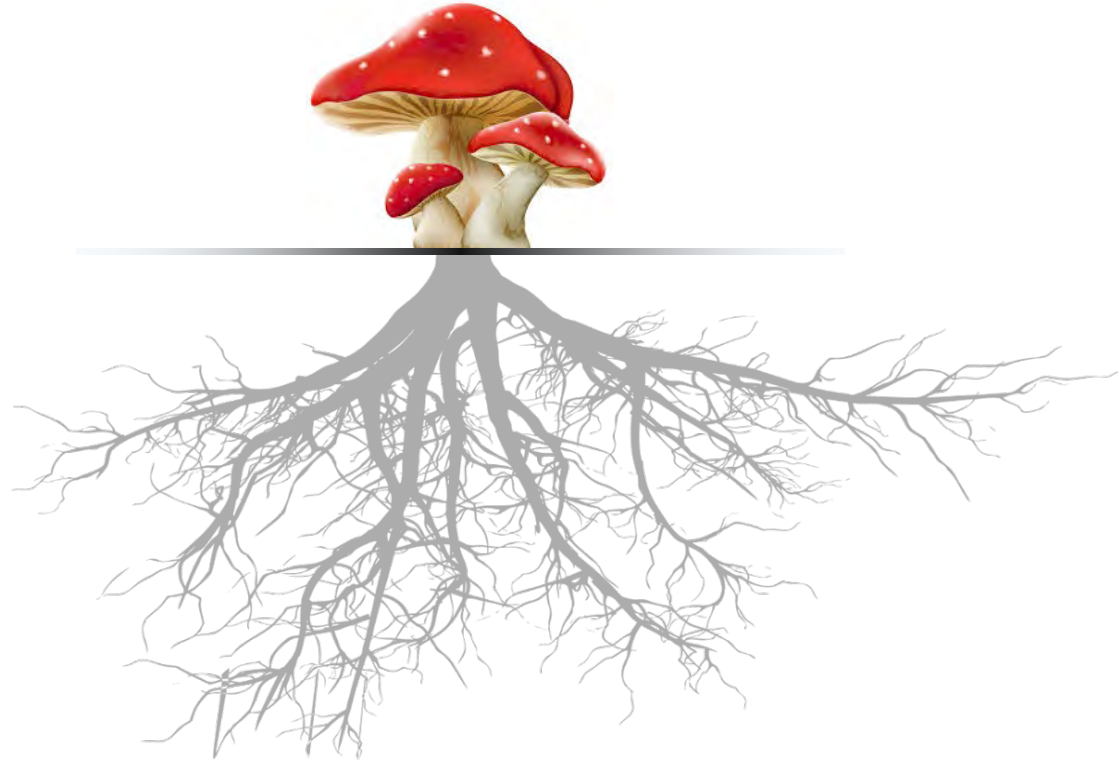
**Compression-only in full 3D?**

## mushroom

(fruit, the "plant")

## mycelium

(roots, the "animal")





# mycelium

the vegetative part of a fungus or fungus-like bacterial colony, consisting of a mass of branching, thread-like hyphae.

## advantages

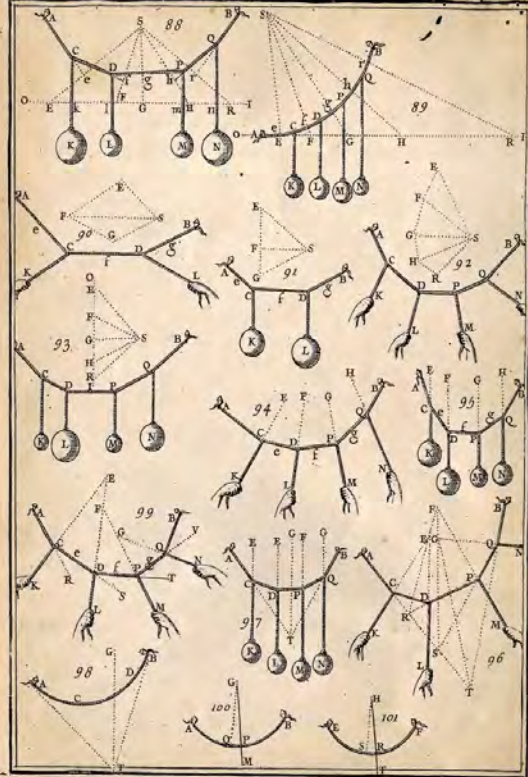
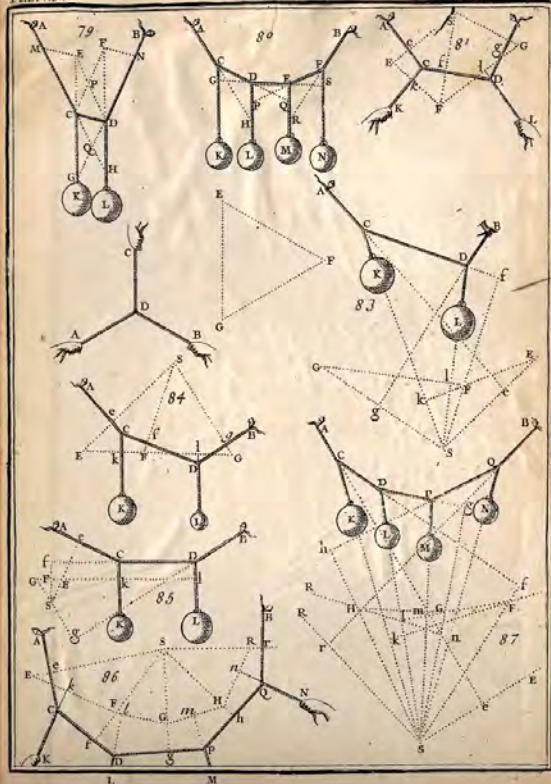
1. Waste as fuel
2. Reverse carbon emissions
3. Fast production
4. Light weight
5. Water repellent
6. Grows anywhere
7. Compostable

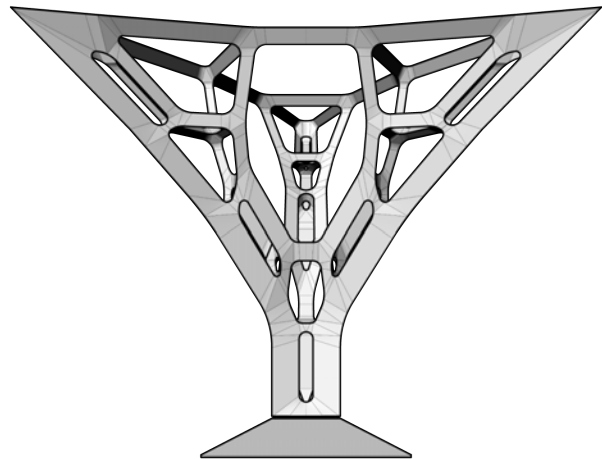
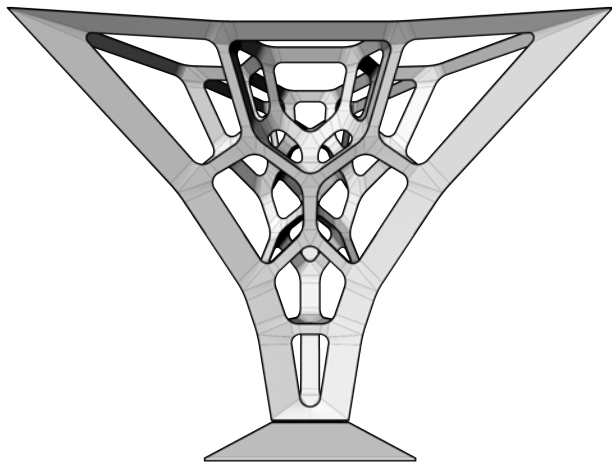
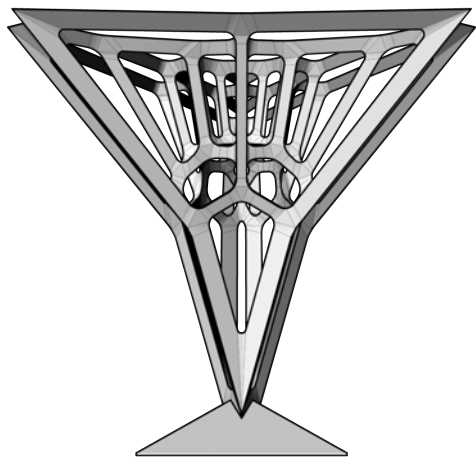


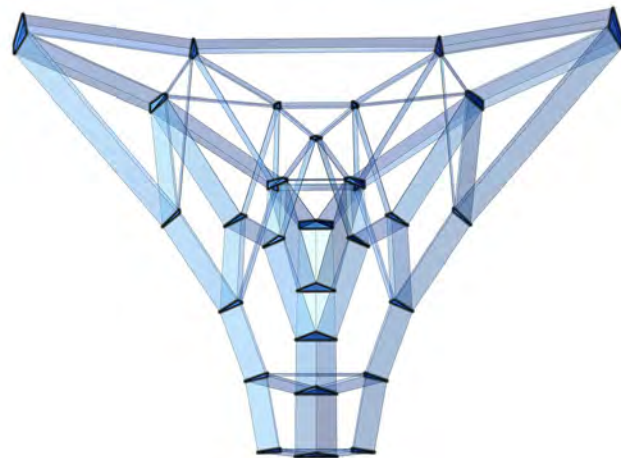
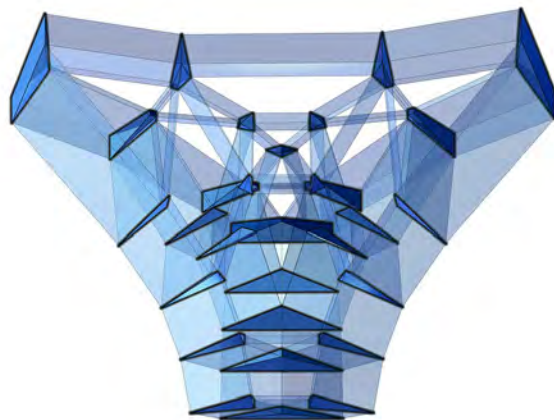
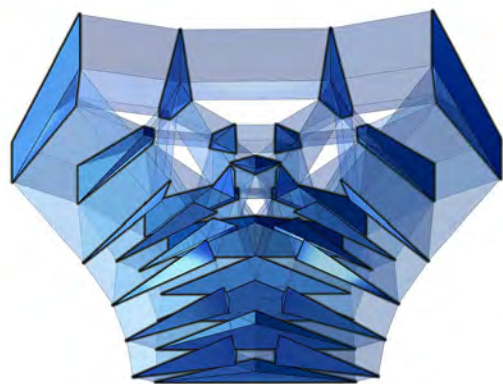
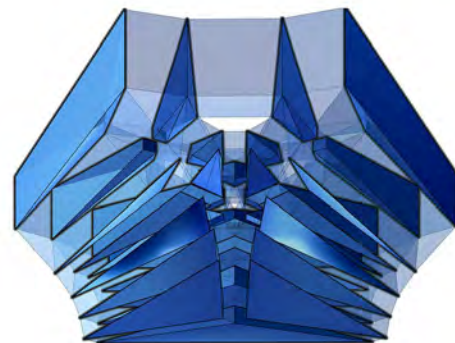
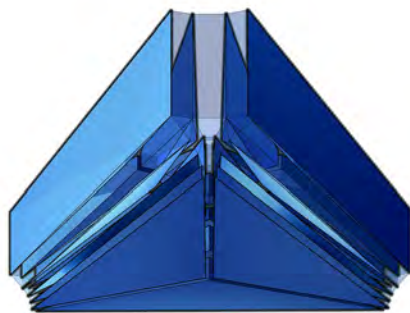
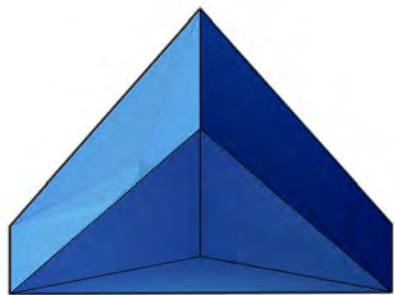


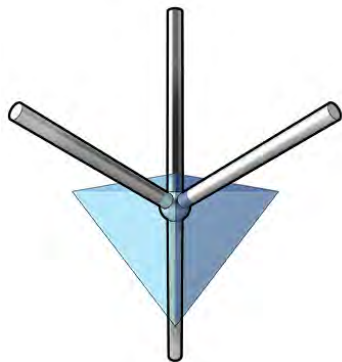
A vertical mechanical testing setup is shown. It consists of a central specimen, a light-colored rectangular block, held between two large, polished metal cylindrical components. Above the specimen, there are three more stacked metal rings. The entire assembly is mounted on a base with a circular hole and several small screws. The base is resting on a dark, ribbed surface. The background is a plain, light-colored wall.

**0.15 – 0.4 MPa**

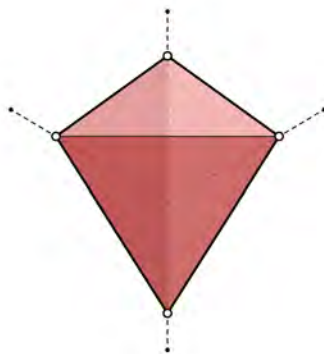




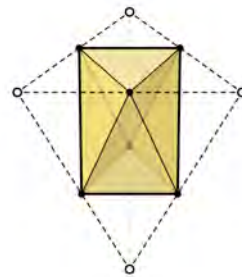




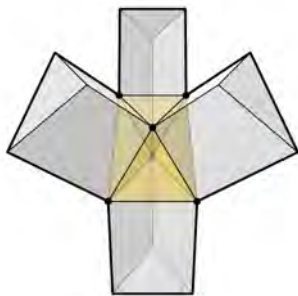
1. force polyhedron



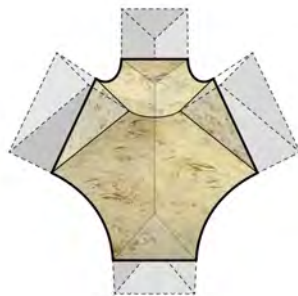
2. EGI



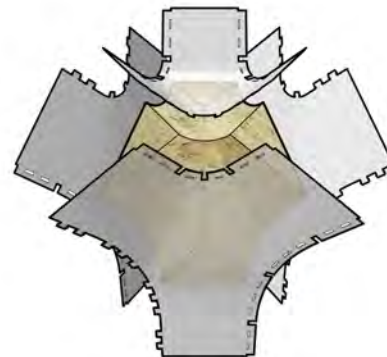
3. EGI dual



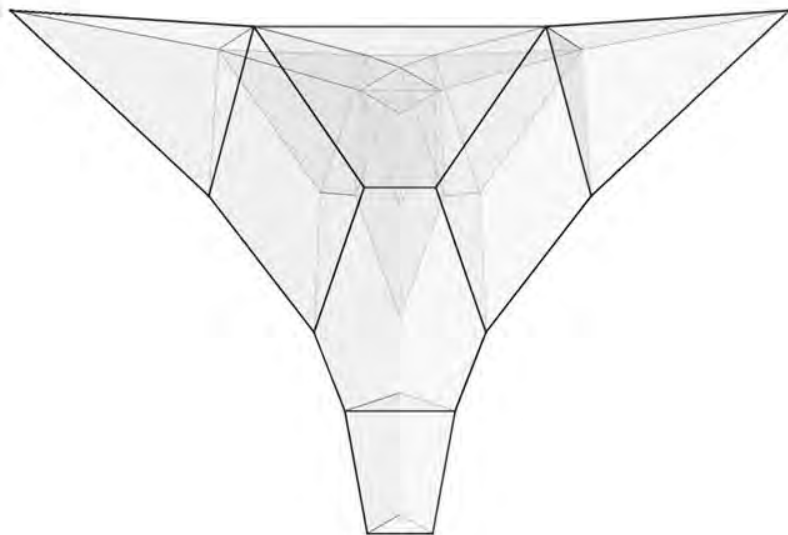
4. massing



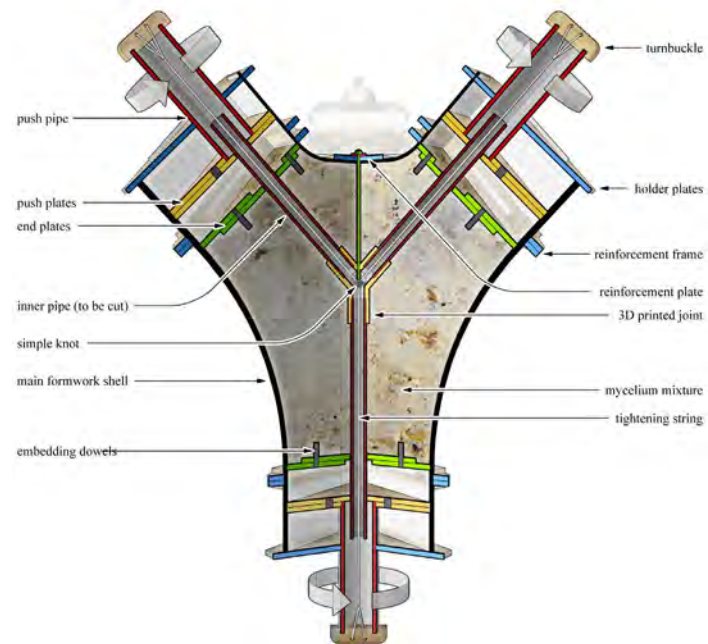
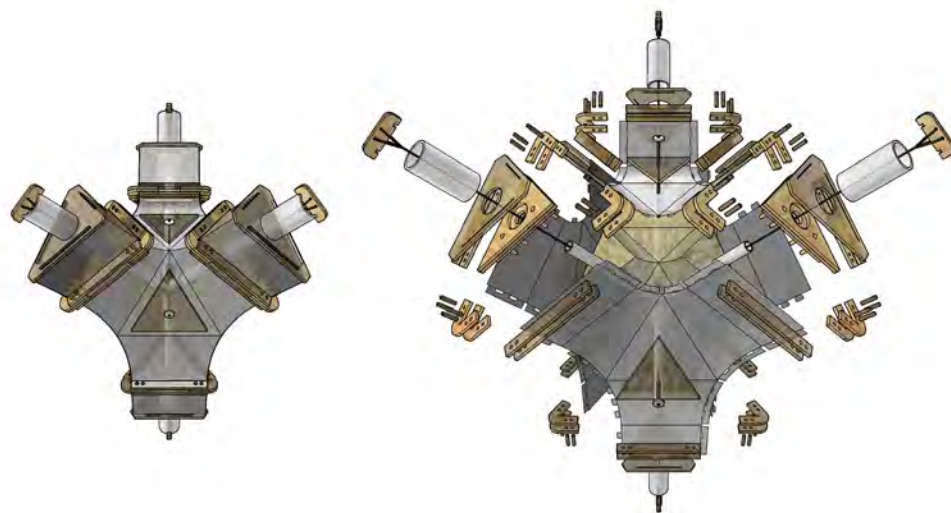
5. target geometry



6. mould geometry







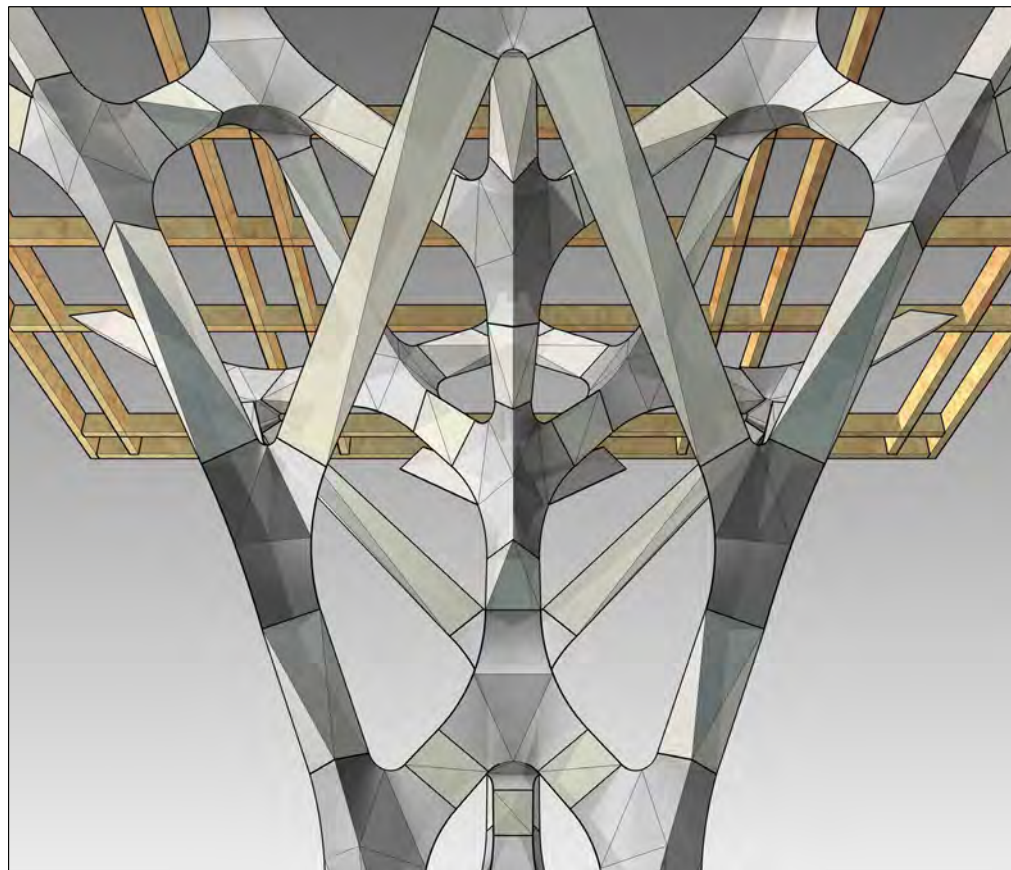


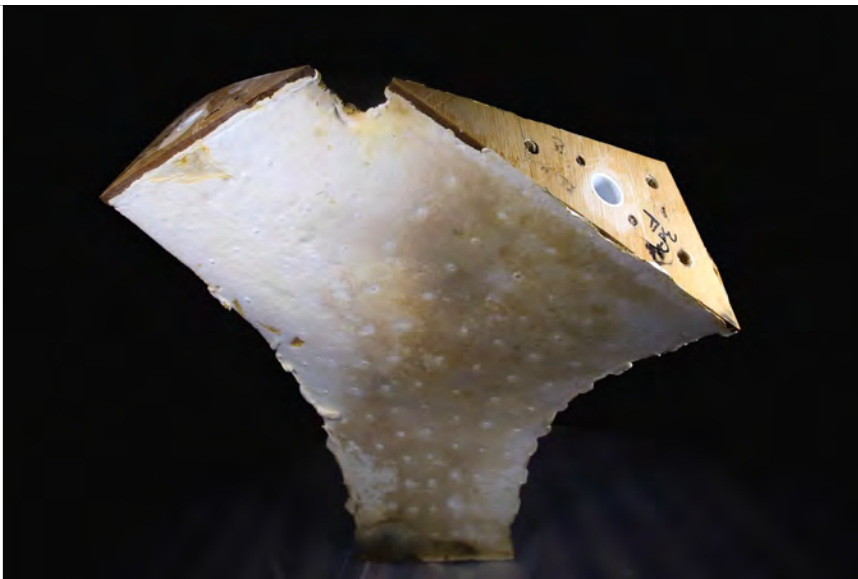
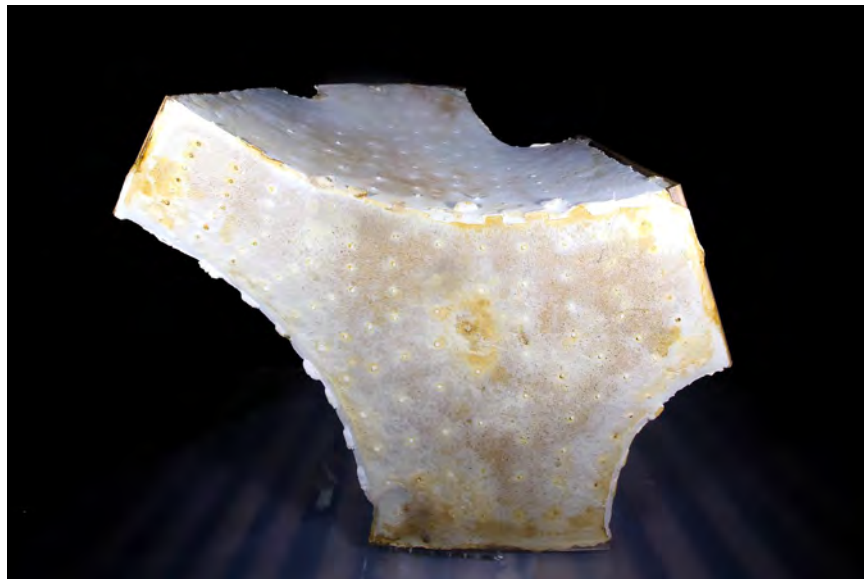


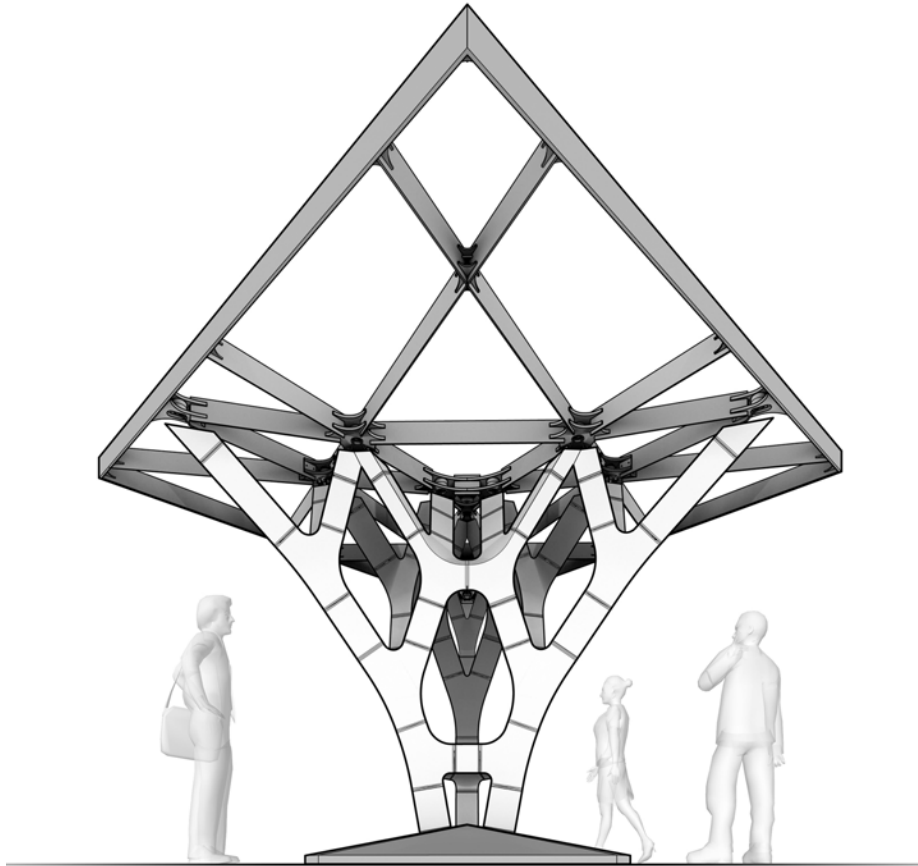
From polyhedral force diagram



From cell network

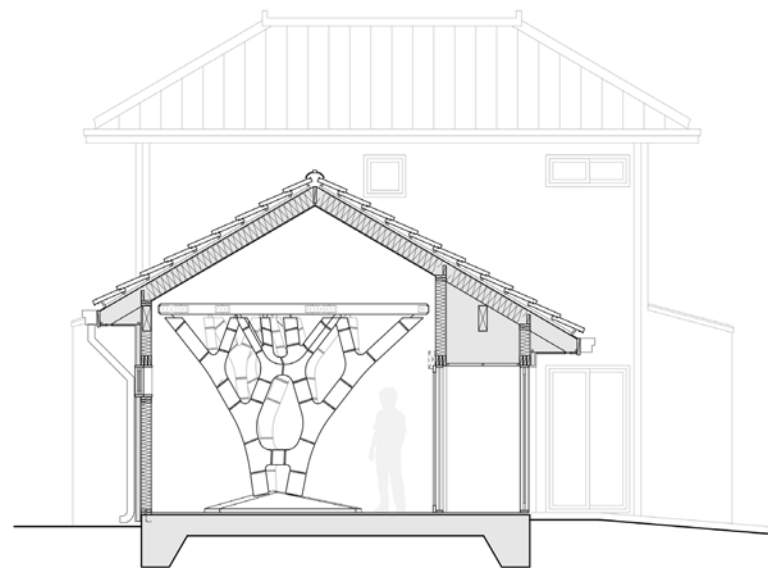
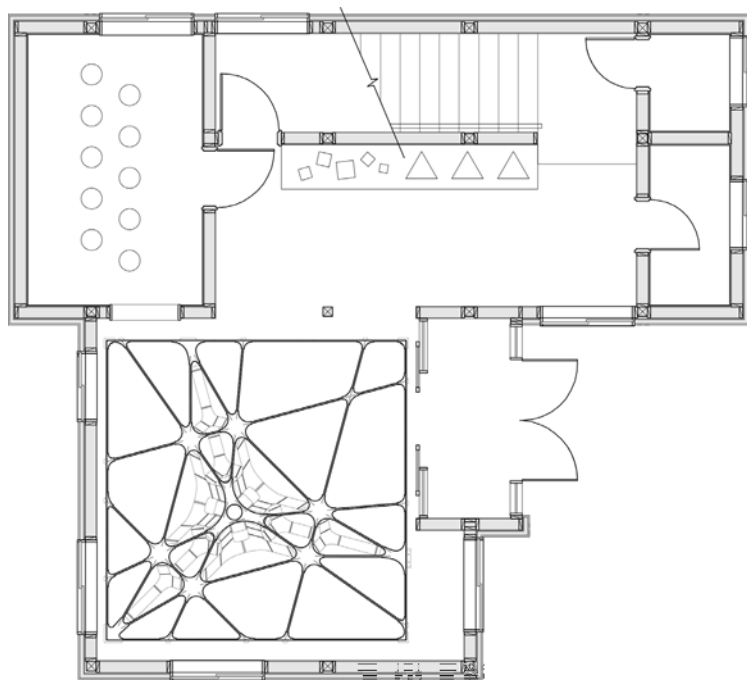






- 4m x 4m x 3m (height)
- Weight
 

|                 |          |
|-----------------|----------|
|                 | 315.9 kg |
| Grid            | 133.7 kg |
| Tree (mycelium) | 182.2 kg |
- 36 linear mycelium members, ranging from 17cm to 60cm in length
- 15 total nodes (5 types), ranging from 5.94 kg to 9.78.8kg
- Mycelium strength: 0.2 Mpa
- Design + development : 6 months
- Construction time : **1 week**



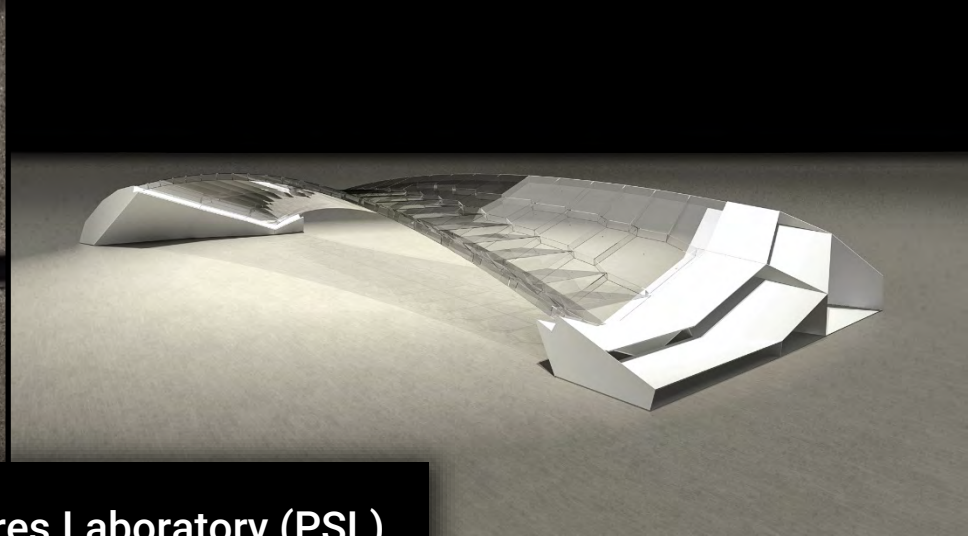
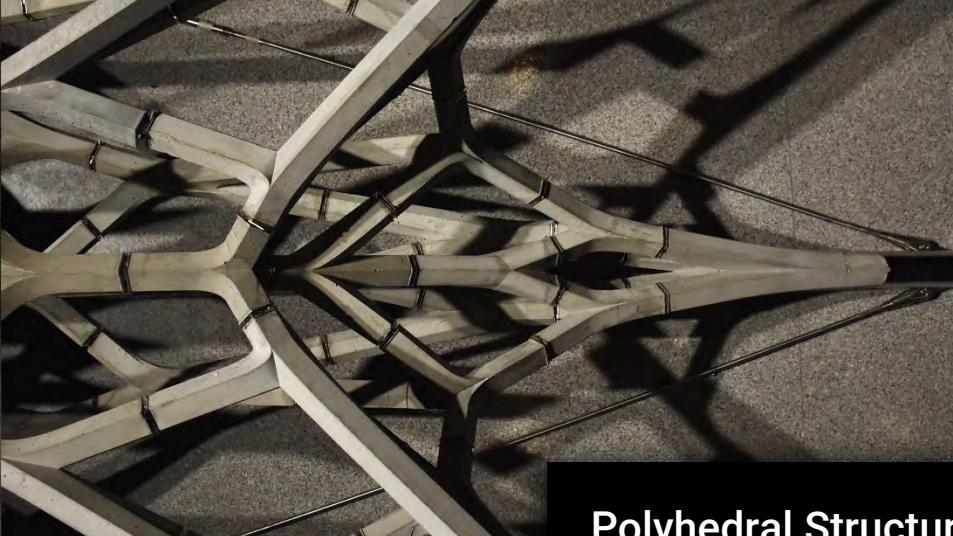






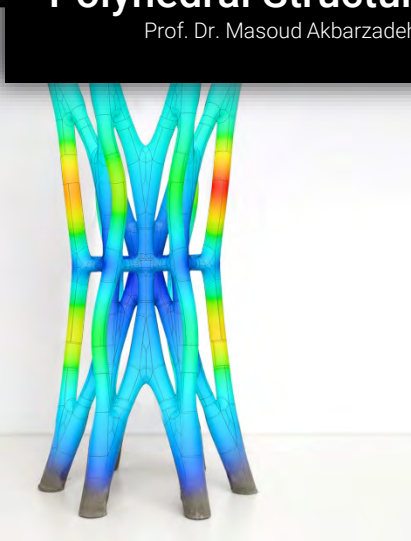


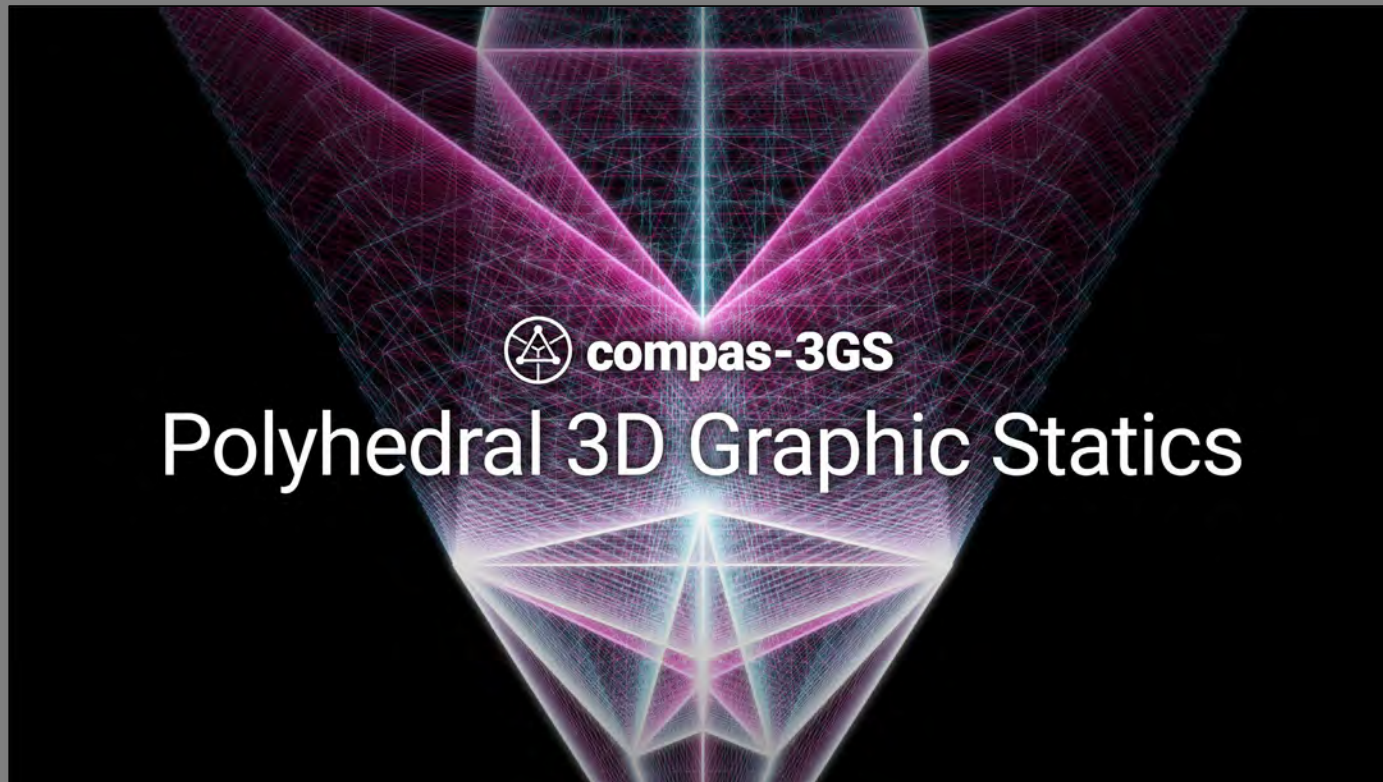




## Polyhedral Structures Laboratory (PSL)

Prof. Dr. Masoud Akbarzadeh • University of Pennsylvania





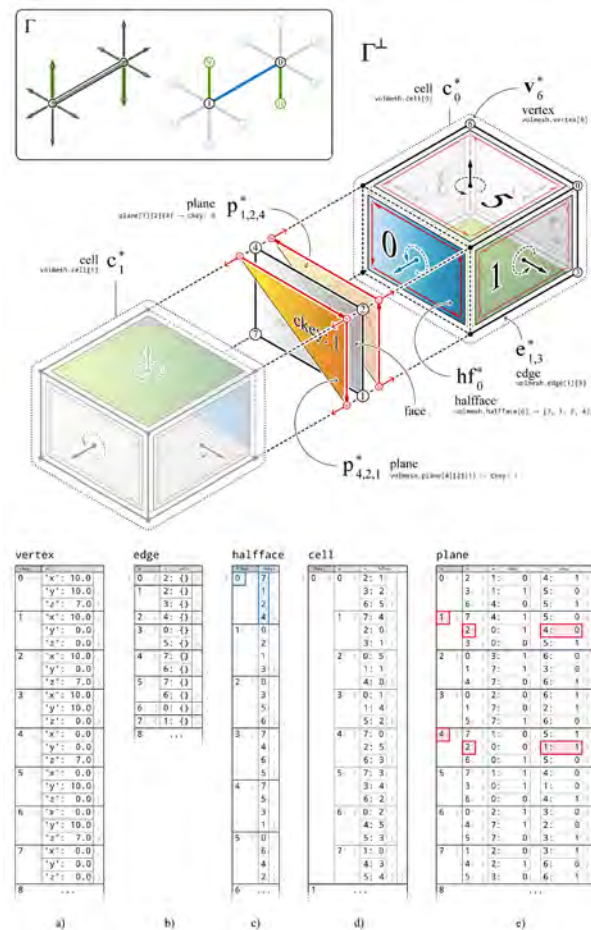
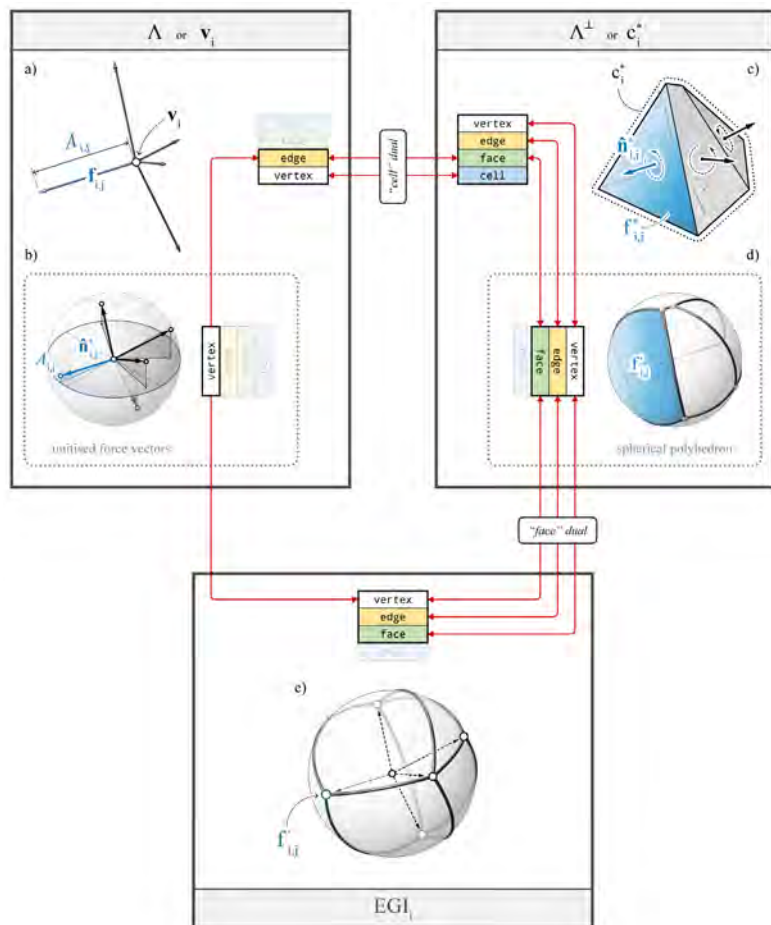
 **compas-3GS**  
Polyhedral 3D Graphic Statics



# C O M P A S

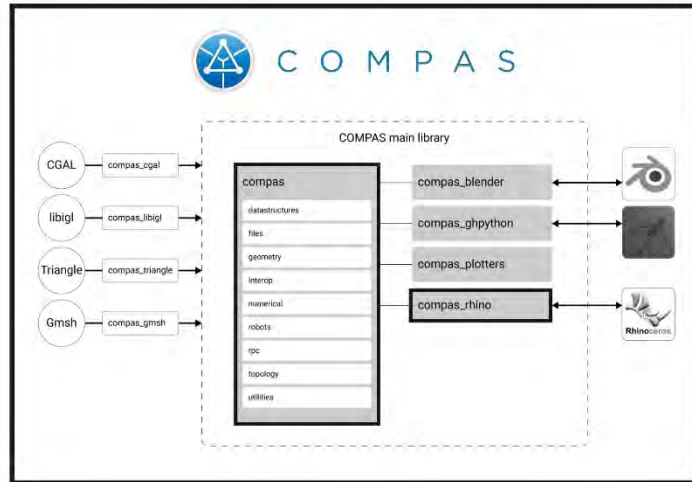
Open-source, Python-based framework for computational research and collaboration  
in architecture, engineering and digital fabrication

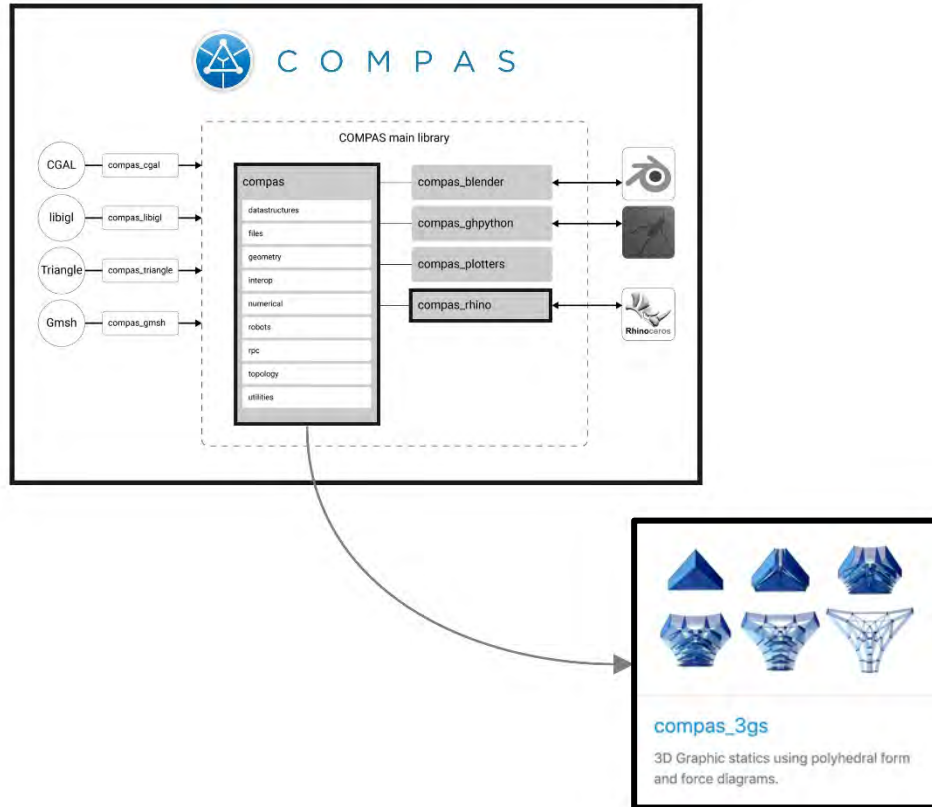
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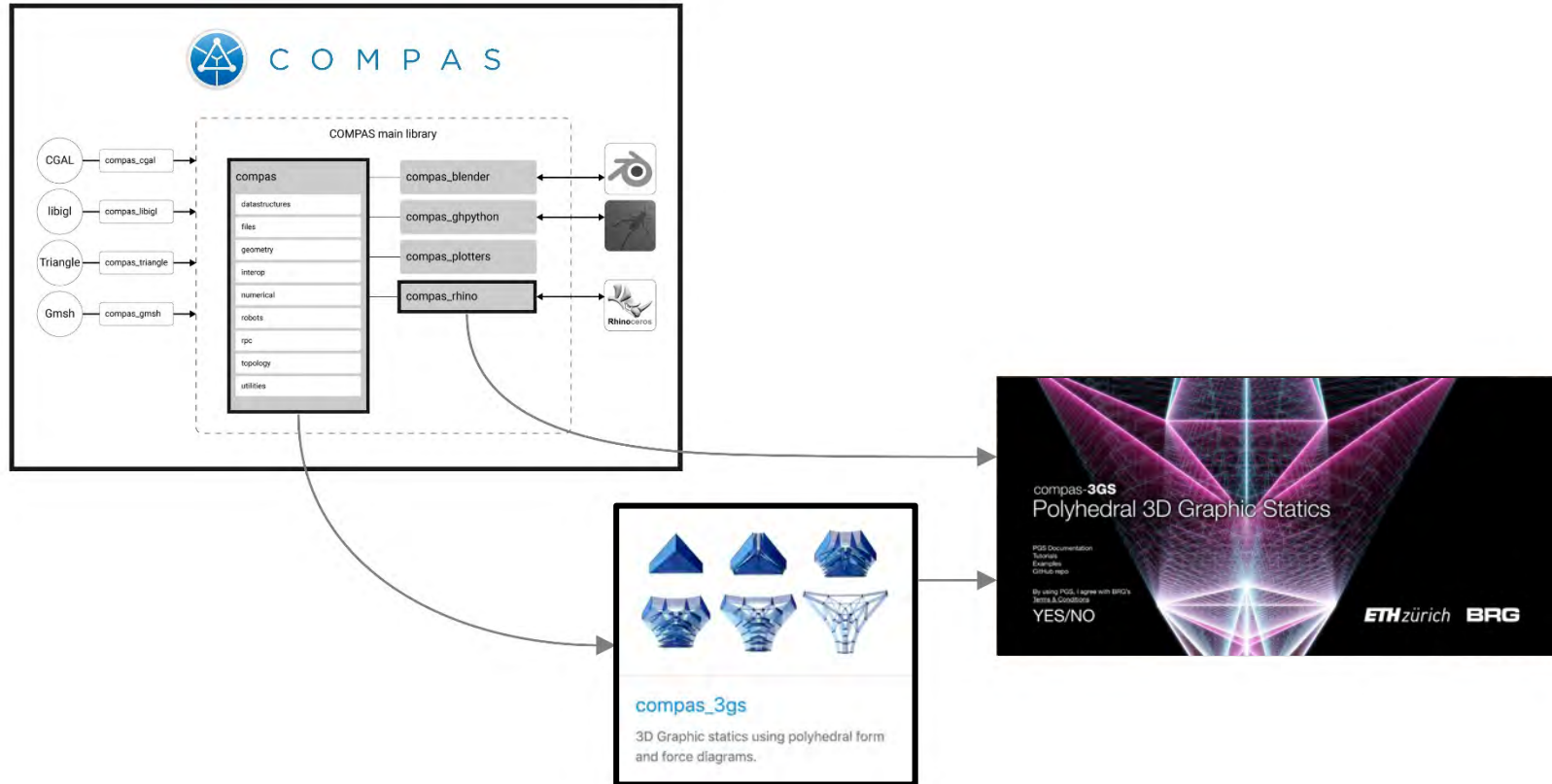


| vertex                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | edge                                                                                                                                                                         | halfedge                                                                                                                                 | cell                                                                                                                                                                                                                                                                                                                                                          | plane                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
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a) b) c) d) e)







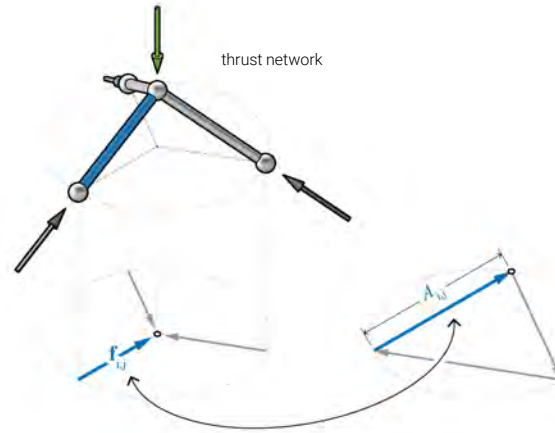
## 2D graphic statics



form diagram

force diagram

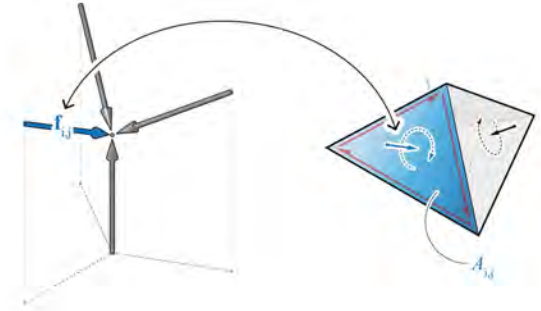
## 2.5D graphic statics



form diagram

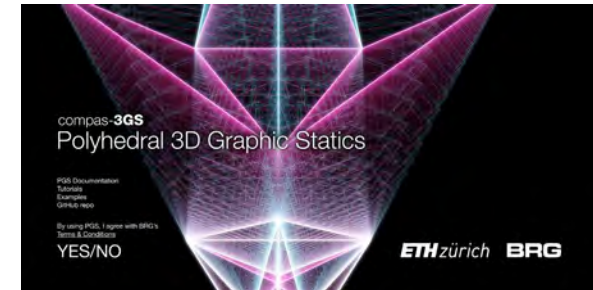
force diagram

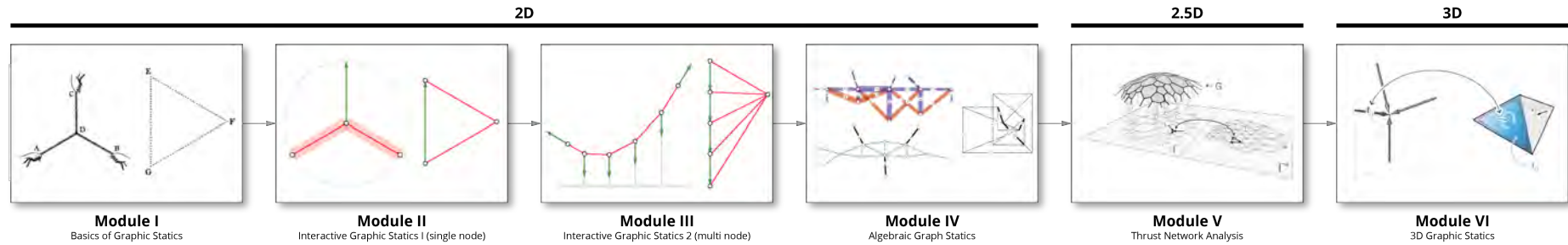
## 3D graphic statics



form diagram

force diagram





A large crowd of LEGO minifigures is seated in yellow stadium-style seats, filling the frame. The minifigures are diverse in appearance, wearing various hats, clothing, and accessories, representing a global population. The text "2.1+ billion more people by 2050" is overlaid in white on the center of the image.

2.1+ billion more people  
by 2050

An aerial night photograph of the Manhattan skyline, showing a dense cluster of illuminated skyscrapers and buildings. The Hudson River is visible on the left, and the East River on the right. The city lights reflect on the water. The text "One Manhattan. Every week. For the next 30 years." is overlaid in white.

One Manhattan. Every week. For the next 30 years.

## Environmental Impact of AEC Industry

Emissions

40%

Improving the rate of resource productivity

—**doing more with less**—

faster than the economic growth rate  
is the notion behind “decoupling”...

That goal, however, demands an **urgent rethink** of  
the links between resource use and economic prosperity,  
buttressed by a massive investment in  
**technological, financial and social innovation.**

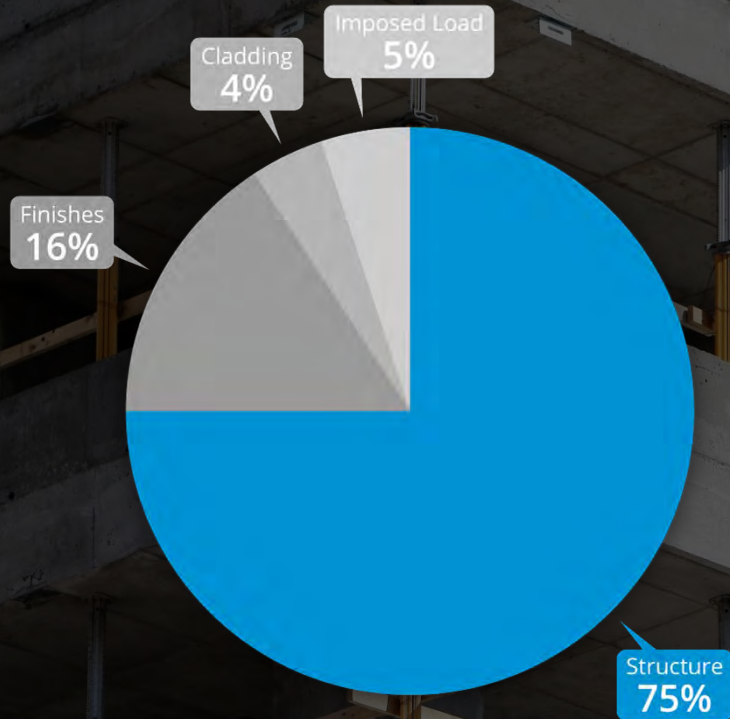
Energy

40%

“Decoupling Natural Resource Use and Environmental Impacts from Economic Growth”

United Nations Environmental Programme (2011)

## Superstructure of a Typical Mid-rise Building





We need to  
change the way  
we **design**  
structures

We need to  
change the way  
we **build**  
structures

# Towards Robotically Fabricated Composite Building Structures

R&D for the BUGA Fibre Pavilion 2019 and the livMatS Pavilion 2021

*Serban Bodea, Block Research Group*



063-0605-00L : Computational Structural Design 1

# Computational Graphic Statics

Dr. Juney Lee & Dr. Lluís Enrique

Autumn Semester 2021

**ETH** zürich

**DARCH**

**41**

**BRG**



compas dashboard

**v0.7.1**



3GS

**V0.2.0**